Online Appendix

Target-Based Communication under Credibility-Distorted Beliefs

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Appendix A. Proofs

A.1. Proposition 1

PROPOSITION 1 (Subjective Credibility). *Under Assumptions 1–4, the probability that cohort g attaches to hypothesis T is*

$$\chi_{g,t} \;=\; \frac{e^{\,b U_{T,g,t-1}}}{e^{\,b U_{T,g,t-1}} + e^{\,b U_{E,g,t-1}}} \;=\; \Big[1 + e^{-b\,(U_{T,g,t-1} - U_{E,g,t-1})}\Big]^{-1}, \qquad b \;=\; \frac{1}{2\sigma_g^2\,(1-\rho_U)}. \tag{8}$$

where the performance $U_{h,t}$ of hypothesis $h \in \{T, E\}$ is

$$U_{h,t} \;=\; -(1-\rho_U)\,e_{h,t}^2 + \rho_U\,U_{h,t-1}. \tag{9}$$

PROOF. Under Equation (3) with anchor m_h, the forecast of π_t made at $t-1$ is $\rho_g \pi_{t-1} + (1-\rho_g) m_h$ and its error is Equation (5), $e_{h,t} = u_t - (1-\rho_g) m_h$ with $u_t \equiv \pi_t - \rho_g \pi_{t-1}$. Since u_t is common to both hypotheses, the two errors differ by a constant,

$$e_{E,t} - e_{T,t} = (1-\rho_g)(m_T - m_{E,g}) = -(1-\rho_g)(1-\nu)\,\Delta_g, \tag{A1}$$

using $m_T - m_{E,g} = (1-\nu)(\pi^* - \bar{\pi}_g)$ from Equation (4).

Under the same law, $\pi_t \mid \pi_{t-1}, h \sim N(\rho_g \pi_{t-1} + (1-\rho_g) m_h, \sigma_g^2)$, whose log-density is Equation (6); by Assumption 6 these are the errors the cohort scores. Substituting

into Equation (7),

$$\ell_{h,t} = c_t - \frac{1}{2\sigma_g^2} \sum_{j\geq 0} \rho_U^j \, e_{h,t-j}^2, \qquad c_t \equiv -\tfrac{1}{2}\log(2\pi\sigma_g^2) \sum_{j\geq 0} \rho_U^j, \tag{A2}$$

where c_t is common to both hypotheses because they share σ_g. Define the performance of hypothesis h as the discounted average of its squared errors,

$$U_{h,t} \equiv -(1-\rho_U) \sum_{j\geq 0} \rho_U^j \, e_{h,t-j}^2, \tag{A3}$$

whose weights $(1-\rho_U)\rho_U^j$ sum to one. Separating the $j=0$ term,

$$U_{h,t} = -(1-\rho_U)\, e_{h,t}^2 + \rho_U \Big[-(1-\rho_U) \sum_{i\geq 0} \rho_U^i \, e_{h,t-1-i}^2 \Big] = -(1-\rho_U)\, e_{h,t}^2 + \rho_U \, U_{h,t-1},$$

which is Equation (9). Comparing (A2) with (A3), the sum in (A2) equals $-U_{h,t}/(1-\rho_U)$, hence

$$\ell_{h,t} = c_t + \frac{U_{h,t}}{2\sigma_g^2(1-\rho_U)}. \tag{A4}$$

Finally, by Assumption 4 the prior is $1/2$ on each hypothesis and the likelihoods are $e^{\ell_{h,t-1}}$, the information being that up to $t-1$ (the timing of Koop and Korobilis (2012) adopted in the text). Bayes' rule gives

$$\chi_{g,t} = \frac{\frac{1}{2}e^{\ell_{T,t-1}}}{\frac{1}{2}e^{\ell_{T,t-1}} + \frac{1}{2}e^{\ell_{E,t-1}}} = \Big[1 + e^{-(\ell_{T,t-1} - \ell_{E,t-1})}\Big]^{-1},$$

and by (A4) the constant c_{t-1} cancels in the difference, $\ell_{T,t-1} - \ell_{E,t-1} = (U_{T,t-1} - U_{E,t-1})/[2\sigma_g^2(1-\rho_U)]$. Naming the factor b delivers Equation (8). By Assumption 1, $\chi_{g,t}$ is the probability held by one agent, not a population share. □

A.2. Lemma 1

LEMMA 1 (Predictive distribution and the posterior-mean anchor). *Under Assumptions 1–4, cohort g's predictive distribution for π_{t+1} is the mixture*

$$\begin{aligned} f(\pi_{t+1} \mid I_t) &= \chi_{g,t}\, \varphi\big(\pi_{t+1}; \mu_{T,t}, \sigma_g^2\big) + (1-\chi_{g,t})\, \varphi\big(\pi_{t+1}; \mu_{E,t}, \sigma_g^2\big), \\ \mu_{h,t} &\equiv \rho_g \pi_t + (1-\rho_g) m_h, \end{aligned} \tag{10}$$

where $\varphi(x;\mu,\sigma^2)$ *is the normal density, and its mean is*

$$\begin{aligned}\mathbb{E}_{g,t}[\pi_{t+1}] &= \rho_g\,\pi_t + (1-\rho_g)\,m_{g,t},\\ m_{g,t} &\equiv \chi_{g,t}\,\pi^* + (1-\chi_{g,t})\,m_{E,g} \;=\; \pi^* + (1-\chi_{g,t})(1-\nu)\,\Delta_g.\end{aligned} \tag{11}$$

where the distance between remembered inflation and the target,

$$\Delta_g \;\equiv\; \bar{\pi}_g - \pi^*,$$

is the cohort's experienced gap.

PROOF. Under Equation (3) with anchor m_h, $\pi_{t+1} \mid I_t, h \sim N(\mu_{h,t}, \sigma_g^2)$ with $\mu_{h,t} = \rho_g\pi_t + (1-\rho_g)m_h$. Not knowing the anchor, the cohort weights the two conditional densities of Assumption 3 by the posterior probabilities of Proposition 1, $\chi_{g,t}$ on T and $1-\chi_{g,t}$ on E, which is the mixture (10); by Assumption 1 the mixing is over hypotheses inside one agent, not over people. Its mean is

$$\mathbb{E}_{g,t}[\pi_{t+1}] = \chi_{g,t}\,\mu_{T,t} + (1-\chi_{g,t})\,\mu_{E,t} = \rho_g\,\pi_t + (1-\rho_g)\big[\chi_{g,t}\,m_T + (1-\chi_{g,t})\,m_{E,g}\big],$$

and substituting Equation (4) in the bracket,

$$m_{g,t} \equiv \chi_{g,t}\,\pi^* + (1-\chi_{g,t})\,m_{E,g} = \pi^* + (1-\chi_{g,t})\,(m_{E,g}-\pi^*) = \pi^* + (1-\chi_{g,t})(1-\nu)\,\Delta_g,$$

which is Equation (11). The same weights applied to the errors (5) give, for any dates s and t,

$$\chi_{g,s}\,e_{T,t} + (1-\chi_{g,s})\,e_{E,t} = u_t - (1-\rho_g)\big[\chi_{g,s}\,m_T + (1-\chi_{g,s})\,m_{E,g}\big] = u_t - (1-\rho_g)\,m_{g,s}, \tag{A5}$$

the error of the forecast made with the anchor $m_{g,s}$; it is used in the proof of Proposition 2. □

A.3. Lemma A1

LEMMA A1 (Diagnostic expectation of a Gaussian forecast). *Let* $f(\cdot \mid I_t)$ *be* $N(\mu_t, \sigma^2)$ *and* $f(\cdot \mid I_{t-1})$ *be* $N(\mu_{t-1}, \sigma^2)$. *Then the diagnostic distribution of Equation* (12) *is normal with the same variance* σ^2 *and mean*

$$\mathbb{E}_t^{\theta}[\pi_{t+1}] \;=\; (1+\theta)\,\mu_t - \theta\,\mu_{t-1} \;=\; \mathbb{E}_t[\pi_{t+1}] + \theta\big(\mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_{t+1}]\big).$$

Moreover, if the cohort's beliefs are about a vector z_t that it believes follows $z_{t+1} = Az_t + \varepsilon_{t+1}$ with Gaussian ε_{t+1}, then

$$\mathbb{E}_t^{\theta}[z_{t+1}] \;=\; (1+\theta)\,A\,z_t - \theta\,A^2 z_{t-1}. \tag{A6}$$

PROOF. Taking logs of Equation (12), the terms in x are

$$-\frac{1}{2\sigma^2}\Big[(x-\mu_t)^2 + \theta\big((x-\mu_t)^2 - (x-\mu_{t-1})^2\big)\Big] = -\frac{1}{2\sigma^2}\Big[(1+\theta)(x-\mu_t)^2 - \theta(x-\mu_{t-1})^2\Big].$$

Completing the square with $m \equiv (1+\theta)\mu_t - \theta\mu_{t-1}$,

$$(1+\theta)(x-\mu_t)^2 - \theta(x-\mu_{t-1})^2 = (x-m)^2 - m^2 + (1+\theta)\mu_t^2 - \theta\mu_{t-1}^2, \tag{A7}$$

and since $m^2 = (1+\theta)^2\mu_t^2 - 2\theta(1+\theta)\mu_t\mu_{t-1} + \theta^2\mu_{t-1}^2$, the constant collects into

$$-m^2 + (1+\theta)\mu_t^2 - \theta\mu_{t-1}^2 = -\theta(1+\theta)\,(\mu_t - \mu_{t-1})^2. \tag{A8}$$

The coefficient on x^2 in (A7) is one, so f^{θ} is normal with variance σ^2 and mean m, which is Equation (13); the constant (A8) does not involve x and is absorbed by the normalisation (it is the mass the operator moves across components in Remark A1).

For the vector case, $\mathbb{E}_t[z_{t+1}] = Az_t$ and $\mathbb{E}_{t-1}[z_{t+1}] = A\,\mathbb{E}_{t-1}[z_t] = A^2 z_{t-1}$. With the reference density carrying the same covariance as the current one, as in the scalar case, Equation (13) applied coordinate by coordinate gives Equation (A6). □

A.4. Proposition 2

PROPOSITION 2 (Diagnostic Expectation with Endogenous Credibility). *FOr cohort g, the Inflation Expectations linearized around the steady state, $= \pi^*$, is*

$$\mathbb{E}_{g,t}^{\theta}[\hat{\pi}_{t+1}] \;=\; \rho_g\,\hat{\pi}_t + (1-\rho_g)\,\kappa_g\,\hat{\chi}_{g,t} + \theta\,\rho_g\big(\hat{\pi}_t - \rho_g\,\hat{\pi}_{t-1} - (1-\rho_g)\,\kappa_g\,\hat{\chi}_{g,t-1}\big). \tag{14}$$

where κ_g is the credibility sensitivity of the anchor,

$$\kappa_g \;=\; -(1-\nu)\,\Delta_g. \tag{15}$$

PROOF. At the steady state $\pi_t = \pi_{t-1} = \pi^*$, $u_t = (1-\rho_g)\pi^*$ and Equation (5) gives

constant errors,

$$e_T = (1-\rho_g)(\pi^* - m_T) = 0, \qquad e_E = (1-\rho_g)(\pi^* - m_{E,g}) = -(1-\rho_g)(1-\nu)\,\Delta_g,$$

so the recursion (9) settles at $U_h = -e_h^2$ and

$$\Delta U \equiv U_T - U_E = (1-\rho_g)^2(1-\nu)^2\,\Delta_g^2 \;\geq\; 0. \tag{A9}$$

Substituting in Equation (8), the rest level of credibility is

$$\chi_g \;=\; \left[1 + e^{-b\,(1-\rho_g)^2(1-\nu)^2\,\Delta_g^2}\right]^{-1} \;\geq\; \tfrac{1}{2}. \tag{A10}$$

where the bound follows from $\Delta U \geq 0$: with inflation at the target, hypothesis T makes no error, so the cohort trusts it at least as much as the experiential hypothesis, and exactly as much when $\nu = 1$ or $\Delta_g = 0$, where the two hypotheses coincide.

Away from the rest point, subtracting the two recursions (9),

$$\Delta U_t = (1-\rho_U)\left(e_{E,t}^2 - e_{T,t}^2\right) + \rho_U\,\Delta U_{t-1}, \tag{A11}$$

and by (A1) together with $e_{E,t} + e_{T,t} = 2u_t - (1-\rho_g)(m_T + m_{E,g})$, the difference of squares is linear in u_t,

$$e_{E,t}^2 - e_{T,t}^2 = (e_{E,t} - e_{T,t})(e_{E,t} + e_{T,t}) = -(1-\rho_g)(1-\nu)\Delta_g\left[2u_t - (1-\rho_g)(m_T + m_{E,g})\right], \tag{A12}$$

which at rest returns (A9). Differentiating (A11) around the rest point, with $du_t = 100\,(\hat{\pi}_t - \rho_g\hat{\pi}_{t-1})$ because u is in percentage points and $\hat{\pi}$ in decimals,

$$d\Delta U_t = -2(1-\rho_U)(1-\rho_g)(1-\nu)\Delta_g\,du_t + \rho_U\,d\Delta U_{t-1}. \tag{A13}$$

Since the logistic $\Lambda(x) = [1+e^{-x}]^{-1}$ has derivative $\Lambda(1-\Lambda)$, Equation (8) gives

$$\hat{\chi}_{g,t} = b\,\chi_g(1-\chi_g)\,d\Delta U_{t-1}. \tag{A14}$$

Subtracting ρ_U times the lag of (A14) and using (A13) at $t-1$,

$$\begin{aligned}\hat{\chi}_{g,t} - \rho_U \hat{\chi}_{g,t-1} &= b\chi_g(1-\chi_g)\big[d\Delta U_{t-1} - \rho_U\, d\Delta U_{t-2}\big] \\ &= -2b\chi_g(1-\chi_g)(1-\rho_U)(1-\rho_g)(1-\nu)\Delta_g\, du_{t-1},\end{aligned}$$

and substituting du_{t-1} delivers the linearised law of credibility,

$$\begin{aligned}\hat{\chi}_{g,t} &= \rho_U\, \hat{\chi}_{g,t-1} - \xi_g\big(\hat{\pi}_{t-1} - \rho_g\, \hat{\pi}_{t-2}\big), \\ \xi_g &= 2b\,\chi_g(1-\chi_g)\,(1-\rho_U)\,(1-\rho_g)\,(1-\nu)\,\Delta_g \times 100.\end{aligned} \tag{A15}$$

Turning to the forecast, Equation (11) gives $\partial m_{g,t}/\partial \chi_{g,t} = -(1-\nu)\Delta_g$ in percentage points, which in decimals is Equation (15). In deviations, the perceived law (3) with the anchor at its posterior mean reads $\hat{\pi}_{t+1} = \rho_g \hat{\pi}_t + (1-\rho_g)\kappa_g \hat{\chi}_{g,t} + \hat{\varepsilon}_{t+1}$, and (A15) shifted one period forward reads $\hat{\chi}_{g,t+1} = \rho_U \hat{\chi}_{g,t} - \xi_g \hat{\pi}_t + \xi_g \rho_g \hat{\pi}_{t-1}$. Stacking $z_{g,t} = (\hat{\pi}_t, \hat{\pi}_{t-1}, \hat{\chi}_{g,t})'$, the cohort perceives the transition

$$z_{g,t+1} \;=\; A_g\, z_{g,t} + \hat{\varepsilon}_{t+1}\, e_1, \qquad A_g \;=\; \begin{pmatrix} \rho_g & 0 & (1-\rho_g)\kappa_g \\ 1 & 0 & 0 \\ -\xi_g & \xi_g \rho_g & \rho_U \end{pmatrix}, \tag{A16}$$

with e_1 the first unit vector, and its one-step forecast is the first row,

$$\mathbb{E}_{g,t}[\hat{\pi}_{t+1}] = e_1' A_g z_{g,t} = \rho_g\, \hat{\pi}_t + (1-\rho_g)\, \kappa_g\, \hat{\chi}_{g,t}. \tag{A17}$$

By Assumption 5 the operator acts on the predictive distribution of this state, and by Assumption 6 its credibility coordinate is the Bayesian $\hat{\chi}_{g,t}$ of (A15), not a distorted one, so Lemma A1 applies with $A = A_g$:

$$\mathbb{E}^{\theta}_{g,t}[\hat{\pi}_{t+1}] = (1+\theta)\, e_1' A_g z_{g,t} - \theta\, e_1' A_g^2 z_{g,t-1}. \tag{A18}$$

Row by row,

$$A_g z_{g,t-1} = \begin{pmatrix} \rho_g \hat{\pi}_{t-1} + (1-\rho_g)\kappa_g \hat{\chi}_{g,t-1} \\ \hat{\pi}_{t-1} \\ \rho_U \hat{\chi}_{g,t-1} - \xi_g \hat{\pi}_{t-1} + \xi_g \rho_g \hat{\pi}_{t-2} \end{pmatrix} = \begin{pmatrix} \mathbb{E}_{g,t-1}[\hat{\pi}_t] \\ \hat{\pi}_{t-1} \\ \hat{\chi}_{g,t} \end{pmatrix},$$

where the third coordinate is (A15) itself: credibility at t is known at $t-1$, so its

$t-1$ forecast is exact. Multiplying by the first row of A_g,

$$e_1' A_g^2 z_{g,t-1} = \rho_g \, \mathbb{E}_{g,t-1}[\hat{\pi}_t] + (1-\rho_g)\kappa_g \, \hat{\chi}_{g,t} = \mathbb{E}_{g,t-1}[\hat{\pi}_{t+1}],$$

and subtracting it from (A17) the credibility terms cancel,

$$\mathbb{E}_{g,t}[\hat{\pi}_{t+1}] - \mathbb{E}_{g,t-1}[\hat{\pi}_{t+1}] = \rho_g\big(\hat{\pi}_t - \mathbb{E}_{g,t-1}[\hat{\pi}_t]\big) = \rho_g\big(\hat{\pi}_t - \rho_g \hat{\pi}_{t-1} - (1-\rho_g)\kappa_g \hat{\chi}_{g,t-1}\big),$$

using (A17) lagged. Substituting into (A18) gives Equation (14). Since (A16) is linear, replacing $\kappa_g \hat{\chi}_{g,t}$ by $m_{g,t}/100$ and undoing the deviations gives the same expression in levels,

$$\mathbb{E}^{\theta}_{g,t}[\pi_{t+1}] \;=\; \rho_g \, \pi_t + (1-\rho_g)\, m_{g,t} + \theta \, \rho_g \big[u_t - (1-\rho_g)\, m_{g,t-1}\big]. \tag{A19}$$

where $m_{g,t}$ is the posterior mean of the anchor of Lemma 1 and $u_t \equiv \pi_t - \rho_g \pi_{t-1}$, whose innovation is, by (A5) with $s = t-1$,

$$u_t - (1-\rho_g)\, m_{g,t-1} = \chi_{g,t-1}\, e_{T,t} + (1-\chi_{g,t-1})\, e_{E,t},$$

the two hypothesis errors weighted by yesterday's credibility. □

REMARK A1 (What the operator gives without Assumptions 5–6). *Equation* (14) *applies Lemma A1 to the* mean *of the mixture* (10)*, with credibility as a predetermined state. The operator* (12) *applied to the cohort's joint distribution of* (π_{t+1}, h)*, with each hypothesis' reference* $\mu_{h,t-1} = \rho_g^2 \pi_{t-1} + (1-\rho_g)(1+\rho_g) m_h$ *and yesterday's weights* $\chi_{h,t-1}$*, gives instead*

$$\begin{aligned} \mathbb{E}^{\theta}_{g,t}[\pi_{t+1}] \;&=\; \rho_g \, \pi_t + (1-\rho_g)\, a^{\theta}_{g,t} + \theta \, \rho_g \big[u_t - (1-\rho_g)\, a^{\theta}_{g,t}\big], \\ a^{\theta}_{g,t} \;&=\; \pi^* + (1-w_{T,t})(1-\nu)\, \Delta_g, \end{aligned} \tag{A20}$$

where each hypothesis is distorted by its own news $\rho_g e_{h,t}$*, and the weights are distorted too, by the constant* (A8)*:*

$$w_{h,t} \;\propto\; \chi_{h,t}^{1+\theta}\, \chi_{h,t-1}^{-\theta} \, \exp\Big\{ \frac{\theta(1+\theta)\, \rho_g^2 \, e_{h,t}^2}{2\sigma_g^2} \Big\}. \tag{A21}$$

With $\theta < 0$ *the weights under-react too: the same* θ *that produces under-reaction in the forecast produces under-reaction in the posterior. Relative to* (A20)*, the model* (14) *omits*

two things: the credibility that enters the innovation is yesterday's, $m_{g,t-1}$, rather than today's, a gap of order $\theta\rho_g(1-\rho_g)(m_{g,t}-m_{g,t-1})$; and the weights are the Bayesian χ rather than w. Both are choices (Assumptions 5–6) made so that the expectation is linear in the state (A16)*, which the HANK's perceived transition needs, and so that* θ *acts only on the forecast while ξ_χ pins down the credibility process. Equation* (A20) *is computable with the level filter of the simple model by replacing χ with w in the anchor and using today's weight in the innovation; it is the robustness check for this choice.*

A.5. Lemma 2

LEMMA 2 (The cohort-g log-linearized consumption). *The first-order log-linearised solution of* (16)–(17) *around $a_g^{ss}=0$, $\pi^{ss}=\pi^*$, χ_g^{ss}, and $r_g^{per\,ss}=\beta_g^{-1}-1$ is*

$$c_{g,t} \;=\; (1-\beta_g)\sum_{h=0}^{\infty}\beta_g^h\, y_{t+h} \;-\; \sigma\sum_{h=0}^{\infty}\beta_g^{h+1}\,\hat{r}^{per}_{g,t+h}, \tag{21}$$

where $\hat{r}^{per}_{g,t+h}$ is given by (20).

PROOF. The cohort-g Euler equation in levels is:

$$C_{g,t}^{-1/\sigma} \;=\; \beta_g(1+r_{g,t}^{\text{per}})\,\mathbb{E}^{\theta}_{g,t}[C_{g,t+1}^{-1/\sigma}].$$

where uppercase letters denote levels and $r_{g,t}^{\text{per}}$ is the level of the perceived rate in (18). Log-linearizing around the steady state $\{C_g^{ss},\ r_g^{per\,ss}=\beta_g^{-1}-1,\ A_g^{ss}=0,\ \pi^{ss}=\pi^*,\ \chi_g^{ss}\}$ and following the paper convention that lowercase letters denote first-order deviations from steady state, so $c_{g,t}\equiv\log(C_{g,t}/C_g^{ss})$, $\hat{r}_{g,t}^{\text{per}}\equiv r_{g,t}^{\text{per}}-r_g^{per\,ss}$, and analogously for y and a, the Euler equation becomes

$$c_{g,t+h+1}-c_{g,t+h}=\sigma\,\hat{r}_{g,t+h}^{\text{per}}. \tag{A22}$$

Forward iterating equation A22 yields

$$c_{g,t+h}=c_{g,t}+\sigma\sum_{k=0}^{h-1}\hat{r}_{g,t+k}^{\text{per}}.$$

The next step is to find a close-form expression for the log-linear version of the intertemporal budget constraint. The log-linearized expression of equation 17 around the steady state is:

$$a_{g,t+1} = \beta_g^{-1} a_{g,t} + y_t - c_{g,t}. \tag{A23}$$

Forward iteration yields

$$\sum_{h=0}^{\infty} \beta_g^{h+1} a_{g,t+h+1} = \sum_{h=0}^{\infty} \beta_g^{h} a_{g,t+h} + \sum_{h=0}^{\infty} \beta_g^{h+1} \left(y_{t+h} - c_{g,t+h} \right).$$

which, after index reordering and cancelling common terms is equivalent to:

$$\lim_{T \to \infty} \beta_g^{T} a_{g,T} - a_{g,t} = \sum_{h=0}^{\infty} \beta_g^{h+1} \left(y_{t+h} - c_{g,t+h} \right).$$

Finally, imposing the transversality condition and recalling the assets deviation at t is $a_{g,t} = 0$, we reach a familiar expression:

$$\sum_{h=0}^{\infty} \beta_g^{h} c_{g,t+h} = \sum_{h=0}^{\infty} \beta_g^{h} y_{t+h}. \tag{A24}$$

Substituting the forward-iterated Euler equation into equation A24 and solving for $c_{g,t}$, we have:

$$\frac{1}{1-\beta_g} c_{g,t} = \sum_{h=0}^{\infty} \beta_g^{h} y_{t+h} - \sigma \sum_{h=0}^{\infty} \beta_g^{h} \sum_{k=0}^{h-1} \hat{r}_{g,t+k}^{\text{per}}. \tag{A25}$$

Now observe that after reversing the order of summation in the last term we get

$$\begin{aligned} \sum_{h=0}^{\infty} \beta_g^{h} \sum_{k=0}^{h-1} \hat{r}_{g,t+k}^{\text{per}} &= \hat{r}_{g,t}^{\text{per}} \left(\beta_g + \beta_g^2 + \beta_g^3 + \cdots \right) + \hat{r}_{g,t+1}^{\text{per}} \left(\beta_g^2 + \beta_g^3 + \cdots \right) + \hat{r}_{g,t+2}^{\text{per}} \left(\beta_g^3 + \beta_g^4 + \cdots \right) + \cdots \\ &= \sum_{k=0}^{\infty} \hat{r}_{g,t+k}^{\text{per}} \sum_{h=k+1}^{\infty} \beta_g^{h} = \sum_{k=0}^{\infty} \frac{\beta_g^{k+1}}{1-\beta_g} \hat{r}_{g,t+k}^{\text{per}}. \end{aligned}$$

Therefore, equation A25 turns into

$$\frac{1}{1-\beta_g} c_{g,t} = \sum_{h=0}^{\infty} \beta_g^{h} y_{t+h} - \sigma \sum_{h=0}^{\infty} \frac{\beta_g^{h+1}}{1-\beta_g} \hat{r}_{g,t+h}^{\text{per}}.$$

Multiplying by $1-\beta_g$ yields

$$c_{g,t} \;=\; (1-\beta_g)\sum_{h=0}^{\infty}\beta_g^h\, y_{t+h} \;-\; \sigma\sum_{h=0}^{\infty}\beta_g^{h+1}\,\hat{r}^{\text{per}}_{g,t+h},$$

which is what we wanted to prove. □

A.6. Proposition 3

PROPOSITION 3 (Consumption Wedges and Aggregate Welfare in RANK). *Consider any sequence of structural shocks that produces equilibrium paths $\{\hat{\imath}_t, \hat{\pi}_t, \hat{\chi}_{g,t}\}_{t\geq 0}$ of the RANK model in equations* (23)–(25). *Define $dc_{g,t}$ as the cohort-g consumption path that satisfies its own distorted Euler equation, and dc^R_t as the* ***rational path****: the consumption path a cohort would choose on the same realised paths with perfect foresight of inflation, so that its perceived real rate is the realised one, $\hat{r}^R_t \equiv \hat{\imath}_t - \hat{\pi}_{t+1}$, and $dc^R_t = -\sigma\sum_{s\geq 0}\hat{r}^R_{t+s}$ is common to all cohorts. Denote $\overline{dc}_t \equiv \sum_g \omega_g\, dc_{g,t}$ as the equilibrium aggregate path. The linearized inflation expectations follows $\mathrm{d}\mathbb{E}^{\theta}_{g,t}[\hat{\pi}_{t+1}] = \Lambda^{\pi}_g\hat{\pi}_t + \Lambda^{\chi}_g\hat{\chi}_{g,t}$ with*

$$\Lambda^{\pi}_g\hat{\pi}_t \;\equiv\; \rho_g\big[(1+\theta)\hat{\pi}_t - \theta\rho_g\hat{\pi}_{t-1}\big], \qquad \Lambda^{\chi}_g\hat{\chi}_{g,t} \;\equiv\; (1-\rho_g)\,\kappa_g\big[\hat{\chi}_{g,t} - \theta\rho_g\hat{\chi}_{g,t-1}\big], \tag{43}$$

Let $A_g \equiv -U''_g(c^R_{g,t}) > 0$ denote cohort g's coefficient of absolute risk aversion at the rational path. Then, the ***cohort-g consumption wedge****, $\mathcal{W}^R_{g,t} \equiv dc_{g,t} - dc^R_t$, is the cohort's cumulated one-step-ahead forecast error and splits exactly into a* ***diagnostic error*** *and a* ***credibility error****,*

$$\mathcal{W}^R_{g,t} \;=\; \sigma\sum_{s\geq 0}\Big(\mathrm{d}\mathbb{E}^{\theta}_{g,t+s}[\hat{\pi}_{t+s+1}] - \hat{\pi}_{t+s+1}\Big) \;=\; \sigma\sum_{s\geq 0}\Big[\underbrace{\Lambda^{\pi}_g\hat{\pi}_{t+s} - \hat{\pi}_{t+s+1}}_{\textit{Diagnostic error}} \;+\; \underbrace{\Lambda^{\chi}_g\hat{\chi}_{g,t+s}}_{\textit{Credibility error}}\Big]. \tag{44}$$

Denote the diagnostic error by $\mathcal{A}_{g,t}$ and the credibility error by $(1-\nu)\mathcal{B}_{g,t}$,

$$\mathcal{A}_{g,t} \;\equiv\; \sigma\sum_{s\geq 0}\big(\Lambda^{\pi}_g\hat{\pi}_{t+s} - \hat{\pi}_{t+s+1}\big), \qquad \mathcal{B}_{g,t} \;\equiv\; -\sigma\sum_{s\geq 0}(1-\rho_g)\,\Delta_g\, X_{g,t+s}, \qquad X_{g,t} \;\equiv\; \hat{\chi}_{g,t} - \theta\rho_g\hat{\chi}_{g,t-1}, \tag{45}$$

so that $\mathcal{W}^R_{g,t} = \mathcal{A}_{g,t} + (1-\nu)\mathcal{B}_{g,t}$, where $\mathcal{A}_{g,t}$ does not depend on the intensity ν. Furthermore, the ***cohort-g welfare loss*** *defined in* (42) *is, up to third-order terms, the utility the cohort loses along its chosen path relative to its rational path, measured in*

units of A_g, and it is the sum of the squared diagnostic error, the cross product of the two errors and the squared credibility error,

$$L^R_{g,t} \;=\; \frac{U_g\big(c^R_{g,t}\big) - U_g\big(c_{g,t}\big)}{A_g} \;=\; \tfrac{1}{2}\mathcal{A}^2_{g,t} \;+\; (1-\nu)\,\mathcal{A}_{g,t}\mathcal{B}_{g,t} \;+\; \tfrac{1}{2}(1-\nu)^2\mathcal{B}^2_{g,t}, \qquad (46)$$

while the ***aggregate welfare loss*** *of* (42) *is the cross-cohort dispersion of the cohort paths plus the squared cumulated forecast error of the consensus, Δ^R_t,*

$$\mathcal{L}^R_t \;=\; \tfrac{1}{2}\,\mathrm{Var}_\omega\big(dc_{g,t}\big) \;+\; \tfrac{1}{2}\big(\Delta^R_t\big)^2, \qquad \Delta^R_t \;\equiv\; \overline{dc}_t - dc^R_t \;=\; \sum_g \omega_g\,\mathcal{W}^R_{g,t}. \qquad (47)$$

PROOF. The first step is to write the cohort-g consumption path. Recall the cohort-g Euler equation,

$$C^{-1/\sigma}_{g,t} \;=\; \beta_g(1+r^{\mathrm{per}}_{g,t})\,\mathbb{E}^\theta_{g,t}[C^{-1/\sigma}_{g,t+1}].$$

Log-linearising around the steady state gives $dc_{g,t} = dc_{g,t+1} - \sigma\,\hat{r}^{\mathrm{per}}_{g,t}$, where the perceived rate is given by (20). By Proposition 2, equation (14) written with the operators of (43), the perceived rate is $\hat{r}^{\mathrm{per}}_{g,t} = \hat{\imath}_t - \Lambda^\pi_g\hat{\pi}_t - \Lambda^\chi_g\hat{\chi}_{g,t}$. Substituting,

$$dc_{g,t} \;=\; dc_{g,t+1} \;-\; \sigma\big(\hat{\imath}_t \;-\; \Lambda^\pi_g\hat{\pi}_t \;-\; \Lambda^\chi_g\hat{\chi}_{g,t}\big).$$

Forward iterating and using $\lim_{s\to\infty} dc_{g,t+s} = 0$ yields

$$dc_{g,t} \;=\; -\sigma\sum_{s\geq 0}\big(\hat{\imath}_{t+s} \;-\; \Lambda^\pi_g\hat{\pi}_{t+s} \;-\; \Lambda^\chi_g\hat{\chi}_{g,t+s}\big). \qquad (A26)$$

The next step is the rational path. It solves the same Euler equation on the same realised paths, but with the realised inflation in place of the forecast, $dc^R_t = dc^R_{t+1} - \sigma(\hat{\imath}_t - \hat{\pi}_{t+1})$. Nothing in this equation depends on the cohort, so forward iterating with the same terminal condition gives, for every g,

$$dc^R_t = -\sigma\sum_{s\geq 0}\big(\hat{\imath}_{t+s} - \hat{\pi}_{t+s+1}\big).$$

Subtracting this expression from (A26), the nominal rate cancels and we get the wedge,

$$\mathcal{W}^{R}_{g,t} \;=\; \sigma \sum_{s\geq 0}\big(\Lambda^{\pi}_{g}\hat{\pi}_{t+s} + \Lambda^{\chi}_{g}\hat{\chi}_{g,t+s} - \hat{\pi}_{t+s+1}\big) \;=\; \sigma \sum_{s\geq 0}\Big(\mathrm{d}\mathbb{E}^{\theta}_{g,t+s}[\hat{\pi}_{t+s+1}] - \hat{\pi}_{t+s+1}\Big),$$

which is the sum of the cohort's one-step-ahead forecast errors. Now split each error in two parts. The first part, $\Lambda^{\pi}_{g}\hat{\pi}_{t+s} - \hat{\pi}_{t+s+1}$, is the error the cohort would make with the credibility channel shut, $\kappa_g = 0$. The second part, $\Lambda^{\chi}_{g}\hat{\chi}_{g,t+s}$, is what credibility adds to the forecast, and it is zero when $\kappa_g = 0$. This gives equation (44). The split is exact because the forecast (14) is linear in the two channels.

For the aggregate path, multiply (A26) by ω_g and sum across cohorts,

$$\begin{aligned}\overline{dc}_t \;\equiv\; \sum_{g}\omega_g\, dc_{g,t} \;&=\; -\sigma\sum_{s\geq 0}\Big(\hat{\imath}_{t+s} \;-\; \underbrace{\textstyle\sum_g \omega_g \Lambda^{\pi}_{g}}_{=\bar{\Lambda}^{\pi}}\hat{\pi}_{t+s} \;-\; \underbrace{\textstyle\sum_g \omega_g \Lambda^{\chi}_{g}\hat{\chi}_{g,t+s}}_{=\overline{\Lambda^{\chi}\hat{\chi}}_{t+s}}\Big) \\ &=\; -\sigma\sum_{s\geq 0}\bar{r}^{\,\mathrm{per}}_{t+s},\end{aligned} \tag{A27}$$

which is the aggregate Euler equation (25). Subtracting the rational path,

$$\Delta^{R}_{t} \equiv \overline{dc}_t - dc^{R}_{t} = \sigma\sum_{s\geq 0}\big(\bar{\Lambda}^{\pi}\hat{\pi}_{t+s} + \overline{\Lambda^{\chi}\hat{\chi}}_{t+s} - \hat{\pi}_{t+s+1}\big) = \sum_{g}\omega_g \mathcal{W}^{R}_{g,t},$$

the sum of the forecast errors of the consensus.

The next step is the welfare loss. Take cohort g's discounted utility evaluated on the realised path,

$$U_g\big(\{c_{g,t+s}\}\big) \;=\; \sum_{s\geq 0}\beta_g^{s}\, u(c_{g,t+s}), \tag{A28}$$

and expand it to second order around the rational path $c^{R}_{g,t}$:

$$U_g(c) \;=\; U_g\big(c^{R}_{g,t}\big) \;+\; U_g'\big(c^{R}_{g,t}\big)\big(c - c^{R}_{g,t}\big) \;+\; \tfrac{1}{2}\,U_g''\big(c^{R}_{g,t}\big)\big(c - c^{R}_{g,t}\big)^2 \;+\; O\big(|c - c^{R}_{g,t}|^3\big).$$

The rational path is the optimum of U_g on the cohort's budget set at the realised prices and income. The path the cohort actually chooses, $c_{g,t}$, satisfies the same budget constraint at the same prices, so $c_{g,t} - c^{R}_{g,t}$ is a feasible deviation and the first-order condition of the rational problem removes the linear term. Define

$A_g \equiv -U_g''(c_{g,t}^R) > 0$ as the cohort-g Arrow–Pratt coefficient of absolute risk aversion. Evaluating the expansion at $c = c_{g,t}$,

$$U_g\big(c_{g,t}^R\big) - U_g\big(c_{g,t}\big) \;=\; \tfrac{1}{2} A_g \left(c_{g,t} - c_{g,t}^R\right)^2 \;+\; O\big(\|dc\|^3\big).$$

Dividing by A_g to remove the cohort-specific curvature, using $c_{g,t} - c_{g,t}^R = dc_{g,t} - dc_t^R = \mathcal{W}_{g,t}^R$ and dropping the third-order term, the utility loss is

$$\frac{U_g\big(c_{g,t}^R\big) - U_g\big(c_{g,t}\big)}{A_g} \;=\; \tfrac{1}{2}\left(\mathcal{W}_{g,t}^R\right)^2 \;=\; L_{g,t}^R,$$

which is the cohort welfare loss of (42). This proves the first equality of (46).

The next step is to write the loss in terms of the two errors. By (43), $\Lambda_g^{\chi}\hat{\chi}_{g,t} = (1-\rho_g)\,\kappa_g\,X_{g,t}$ with $X_{g,t} \equiv \hat{\chi}_{g,t} - \theta\rho_g\hat{\chi}_{g,t-1}$, and by (15), $\kappa_g = -(1-\nu)\Delta_g$. Substituting, the credibility error of (44) is

$$\sigma \sum_{s\geq 0} \Lambda_g^{\chi}\hat{\chi}_{g,t+s} = -(1-\nu)\,\sigma \sum_{s\geq 0} (1-\rho_g)\,\Delta_g\,X_{g,t+s} = (1-\nu)\,\mathcal{B}_{g,t},$$

so $\mathcal{W}_{g,t}^R = \mathcal{A}_{g,t} + (1-\nu)\mathcal{B}_{g,t}$ with $\mathcal{A}_{g,t}$ and $\mathcal{B}_{g,t}$ as in (45). $\mathcal{A}_{g,t}$ does not depend on ν because Λ_g^{π} only depends on (ρ_g, θ), and ν moves neither of them (Assumption 2). Expanding the square,

$$\tfrac{1}{2}\big(\mathcal{W}_{g,t}^R\big)^2 = \tfrac{1}{2}\mathcal{A}_{g,t}^2 + (1-\nu)\,\mathcal{A}_{g,t}\mathcal{B}_{g,t} + \tfrac{1}{2}(1-\nu)^2\mathcal{B}_{g,t}^2,$$

which is the second equality of (46). Note that the two errors enter the loss through their squares and through their cross product, so a credibility error of the opposite sign to the diagnostic error lowers the loss.

Finally, the aggregate welfare loss of (42) is $\mathcal{L}_t^R = \sum_g \omega_g\,L_{g,t}^R$. Writing $\mathcal{W}_{g,t}^R = (dc_{g,t} - \overline{dc}_t) + \Delta_t^R$, where Δ_t^R is the same for every cohort,

$$\begin{aligned}\mathcal{L}_t^R \;\equiv\; \sum_g \omega_g\,L_{g,t}^R \;&=\; \tfrac{1}{2}\sum_g \omega_g\big(dc_{g,t} - \overline{dc}_t\big)^2 \;+\; \Delta_t^R \sum_g \omega_g\big(dc_{g,t} - \overline{dc}_t\big) \;+\; \tfrac{1}{2}\big(\Delta_t^R\big)^2 \\ &=\; \tfrac{1}{2}\,\mathrm{Var}_{\omega}\big(dc_{g,t}\big) \;+\; \tfrac{1}{2}\big(\Delta_t^R\big)^2.\end{aligned}$$

The cross term is zero because $\sum_g \omega_g(dc_{g,t} - \overline{dc}_t) = 0$. This is equation (47), which is what we wanted to prove. □

A.7. Proposition 5

PROPOSITION 5 (Optimal Target Communication). *Consider a cost-push shock with realised, already observed, equilibrium paths $\{\hat{\pi}_t, \hat{\chi}_{g,t}\}_{t\geq 0}$ of inflation and cohort credibility.*

a. RANK Model. *Along the realised paths the cohort-g wedge of Proposition 3 is linear in the intensity, $\mathcal{W}^R_{g,t}(\nu) = \mathcal{A}_{g,t} + (1-\nu)\,\mathcal{B}_{g,t}$, with $\mathcal{A}_{g,t}$ and $\mathcal{B}_{g,t}$ the diagnostic and credibility errors of* (45)*: $\mathcal{A}_{g,t}$ does not depend on ν and the credibility error $(1-\nu)\mathcal{B}_{g,t}$ scales with the distance from full intensity. Let the cumulated losses be $L^R_g(\nu) = \frac{1}{2}\sum_{t\geq 0}\big(\mathcal{W}^R_{g,t}(\nu)\big)^2$ and $\mathcal{L}^R(\nu) = \sum_g \omega_g L^R_g(\nu)$. Then the intensity cohort g would choose and the aggregate optimum are*

$$\nu^\star_g \equiv \arg\min_{\nu\in\mathbb{R}} L^R_g(\nu) = 1 + \frac{\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t}}{\sum_{t\geq 0}\mathcal{B}^2_{g,t}},$$
$$\nu^\star_{\text{agg}} \equiv \arg\min_{\nu\in\mathbb{R}} \mathcal{L}^R(\nu) = 1 + \frac{\sum_g \omega_g \sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t}}{\sum_g \omega_g \sum_{t\geq 0}\mathcal{B}^2_{g,t}},$$

whenever the denominators are positive. Full intensity, $\nu^\star = 1$, is optimal only when the two error paths are orthogonal, $\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t} = 0$. The optimum is interior, $\nu^\star < 1$, when the cross product is negative, $\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t} < 0$, that is, when the two errors have opposite signs in the quarters where they are large, so that the credibility error partly offsets the diagnostic one. The ratio $\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t}/\sum_{t\geq 0}\mathcal{B}^2_{g,t}$ is the slope of a regression of $-\mathcal{A}_{g,t}$ on $\mathcal{B}_{g,t}$ through the origin. The cohort optima do not coincide in general.

b. HANK Model. *Along the realised paths the cohort-g wedge of Proposition 4 is linear in the intensity, $\mathcal{W}^{H,R}_{g,t}(\nu) = \mathcal{A}^H_{g,t} + (1-\nu)\mathcal{B}^H_{g,t}$, with*

$$\mathcal{A}^H_{g,t} \equiv \Big[\mathcal{J}^{c,\pi}_g\big(\Lambda^\pi_g - \mathbb{F}\big)\mathrm{d}\boldsymbol{\pi}\Big]_t,$$
$$\mathcal{B}^H_{g,t} \equiv -(1-\rho_g)\,\Delta_g\left[\mathcal{J}^{c,\pi}_g \mathbf{X}_g\right]_t, \qquad \mathbf{X}_g \equiv (\mathbb{I} - \theta\rho_g L)\,\mathrm{d}\boldsymbol{\chi}_g.$$

With the same cumulated losses, $\nu^\star_g$ and $\nu^\star_{\text{agg}}$ take the same form with $(\mathcal{A}^H, \mathcal{B}^H)$ in place of $(\mathcal{A}, \mathcal{B})$, as in the RANK model.

PROOF. *Part 1 (RANK).* The first step is to write the wedge as a function of ν. By Proposition 3, $\mathcal{W}^R_{g,t}(\nu) = \mathcal{A}_{g,t} + (1-\nu)\mathcal{B}_{g,t}$ with $\mathcal{A}_{g,t}$ and $\mathcal{B}_{g,t}$ as in (45). The realised paths are given, so $\mathcal{A}_{g,t}$ and $\mathcal{B}_{g,t}$ are constants in the problem, and $\mathcal{A}_{g,t}$ does not depend on ν.

Cohort problem. Summing the cohort loss (46) over t,

$$L_g^R(\nu) = \tfrac{1}{2}\Big[\sum_{t\geq 0}\mathcal{A}_{g,t}^2 + 2(1-\nu)\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t} + (1-\nu)^2\sum_{t\geq 0}\mathcal{B}_{g,t}^2\Big].$$

This is a quadratic in ν with second derivative $\sum_{t\geq 0}\mathcal{B}_{g,t}^2 > 0$, so its minimum is unique. The first-order condition is

$$-\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t} - (1-\nu)\sum_{t\geq 0}\mathcal{B}_{g,t}^2 = 0,$$

and solving for ν gives $\nu_g^\star = 1 + \sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t}/\sum_{t\geq 0}\mathcal{B}_{g,t}^2$. The minimum loss is

$$\tfrac{1}{2}\Big[\sum_{t\geq 0}\mathcal{A}_{g,t}^2 - \frac{\big(\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t}\big)^2}{\sum_{t\geq 0}\mathcal{B}_{g,t}^2}\Big] \geq 0.$$

That is, the intensity removes the projection of the diagnostic error on the credibility error, and nothing else. From the formula, $\nu_g^\star = 1$ if and only if the two error paths are orthogonal, $\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t} = 0$, and $\nu_g^\star < 1$ if and only if $\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t} < 0$. If $\sum_{t\geq 0}\mathcal{B}_{g,t}^2 = 0$ the loss is $\frac{1}{2}\sum_{t\geq 0}\mathcal{A}_{g,t}^2$ for every ν, and $\nu_g^\star$ is not identified.

Aggregate problem. Multiply the cohort loss by ω_g and sum across cohorts,

$$\mathcal{L}^R(\nu) = \tfrac{1}{2}\sum_g \omega_g\Big[\sum_{t\geq 0}\mathcal{A}_{g,t}^2 + 2(1-\nu)\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t} + (1-\nu)^2\sum_{t\geq 0}\mathcal{B}_{g,t}^2\Big].$$

The first-order condition gives

$$1-\nu_{\text{agg}}^\star = -\frac{\sum_g \omega_g \sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t}}{\sum_g \omega_g \sum_{t\geq 0}\mathcal{B}_{g,t}^2},$$

and the minimum is unique whenever some $\sum_{t\geq 0}\mathcal{B}_{g,t}^2 > 0$. The cohort optimum depends on g through the ratio $\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t}/\sum_{t\geq 0}\mathcal{B}_{g,t}^2$. So the cohort and the aggregate optima coincide only when that ratio is the same for every cohort, for example when $\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t} = 0$ for all g. Otherwise the common intensity is a weighted average of the cohort optima, with weights $\omega_g\sum_{t\geq 0}\mathcal{B}_{g,t}^2$. This proves part 1.

Part 2 (HANK). Fix the realised paths $\mathrm{d}\boldsymbol{\pi}$ and $\mathrm{d}\boldsymbol{\chi}_g$ of the cost-push shock. By (48) the wedge is $[\mathcal{J}_g^{c,\pi}(\Lambda_g^\pi - \mathbb{F})\mathrm{d}\boldsymbol{\pi}]_t = \mathcal{A}_{g,t}^H$ plus $[\mathcal{J}_g^{c,\pi}\Lambda_g^\chi \mathrm{d}\boldsymbol{\chi}_g]_t$. By (43) stacked over horizons, $\Lambda_g^\chi \mathrm{d}\boldsymbol{\chi}_g = (1-\rho_g)\kappa_g \mathbf{X}_g$ with $\mathbf{X}_g = (\mathbb{I} - \theta\rho_g L)\mathrm{d}\boldsymbol{\chi}_g$. Substituting $\kappa_g = -(1-$

$\nu)\Delta_g$,

$$\left[\mathcal{J}_g^{c,\pi}\Lambda_g^{\chi}\mathrm{d}\chi_g\right]_t = -(1-\nu)\,(1-\rho_g)\Delta_g\left[\mathcal{J}_g^{c,\pi}\mathbf{X}_g\right]_t = (1-\nu)\,\mathcal{B}_{g,t}^{H}.$$

The response $[\mathcal{J}_g^{c,\pi}\mathbf{X}_g]_t$ and $\mathcal{A}_{g,t}^{H}$ do not depend on ν, so $\mathcal{W}_{g,t}^{H,R}(\nu) = \mathcal{A}_{g,t}^{H}+(1-\nu)\mathcal{B}_{g,t}^{H}$, the same form as in part 1 with each error weighted by the cohort's own Jacobian. From here the cohort and the aggregate problems are the same quadratics as in part 1 with $(\mathcal{A}^H, \mathcal{B}^H)$ in place of $(\mathcal{A}, \mathcal{B})$, and the same steps give the two optima. Note that the Jacobians enter the optima through the time sums, so the HANK optimum differs from the RANK one even on the same inflation and credibility paths. When the per-capita Jacobian is the same for all cohorts and equal to the permanent-income response, $[\mathcal{J}_g^{c,\pi}\mathbf{x}]_t = \sigma\sum_{s\geq 0} x_{t+s}$, we have $\mathcal{A}_{g,t}^{H} = \mathcal{A}_{g,t}$ and $\mathcal{B}_{g,t}^{H} = \mathcal{B}_{g,t}$, and part 2 reduces to part 1.

Finally, $\kappa_g(1) = 0$ makes $\Lambda_g^{\chi} = 0$ for every cohort in both models. So at $\nu = 1$ the credibility error is zero for any credibility path, realised or not, and the loss is $\frac{1}{2}\sum_{t\geq 0}\mathcal{A}_{g,t}^2$. The diagnostic error is out of reach of target communication, which is what we wanted to prove. □

A.8. Corollary A1

COROLLARY A1 (Who Gains from Target Communication). *Under the conditions of Proposition 5, write the two terms of the cohort-g wedge of Proposition 3 as the diagnostic wedge* $\mathcal{W}_{g,t}^{\pi} \equiv \sigma\sum_{s\geq 0}(\Lambda_g^{\pi} - \bar{\Lambda}^{\pi})\hat{\pi}_{t+s}$ *and the credibility wedge in the absence of communication* $\mathcal{W}_{g,t}^{\chi} \equiv \mathcal{W}_{g,t}^{ex\text{-}post,\chi}(0) = -\sigma\sum_{s\geq 0} D_{g,t+s}$, *with* $D_{g,t}$ *the demeaned credibility exposure defined in the proof of Proposition 5, so that the ex-post wedge at intensity* ν *is* $\mathcal{W}_{g,t}(\nu) = \mathcal{W}_{g,t}^{\pi} + (1-\nu)\mathcal{W}_{g,t}^{\chi}$. *Then the ex-post gain of cohort g from maximal intensity is*

$$L_{g,t}(0) - L_{g,t}(1) = \tfrac{1}{2}\big(\mathcal{W}_{g,t}^{\chi}\big)^2 + \mathcal{W}_{g,t}^{\pi}\,\mathcal{W}_{g,t}^{\chi},$$

which is positive whenever the diagnostic and the credibility wedges of the cohort have the same sign, and negative only if they have opposite signs and $|\mathcal{W}_{g,t}^{\chi}| < 2\,|\mathcal{W}_{g,t}^{\pi}|$. *Hence, maximal intensity is a Pareto improvement over silence whenever the two wedges comove for every cohort, and the share of the cohort's cumulative loss it removes is*

$$\frac{\sum_t\big[\frac{1}{2}(\mathcal{W}_{g,t}^{\chi})^2 + \mathcal{W}_{g,t}^{\pi}\mathcal{W}_{g,t}^{\chi}\big]}{\frac{1}{2}\sum_t\big(\mathcal{W}_{g,t}^{\pi} + \mathcal{W}_{g,t}^{\chi}\big)^2}.$$

PROOF. By Proposition 3 and the scaling step in the proof of part 1 of Proposition 5, the ex-post wedge at intensity ν is $\mathcal{W}_{g,t}(\nu) = \mathcal{W}^{\pi}_{g,t} + (1-\nu)\mathcal{W}^{\chi}_{g,t}$, with $\mathcal{W}^{\pi}_{g,t}$ invariant to ν because Λ^{π}_{g} depends only on (ρ_g, θ). Therefore

$$\begin{aligned} L_{g,t}(0) - L_{g,t}(1) &= \tfrac{1}{2}\big(\mathcal{W}^{\pi}_{g,t} + \mathcal{W}^{\chi}_{g,t}\big)^2 - \tfrac{1}{2}\big(\mathcal{W}^{\pi}_{g,t}\big)^2 \\ &= \tfrac{1}{2}\big(\mathcal{W}^{\chi}_{g,t}\big)^2 + \mathcal{W}^{\pi}_{g,t}\mathcal{W}^{\chi}_{g,t} = \tfrac{1}{2}\,\mathcal{W}^{\chi}_{g,t}\big(\mathcal{W}^{\chi}_{g,t} + 2\mathcal{W}^{\pi}_{g,t}\big). \end{aligned}$$

The product is positive when both wedges have the same sign, since then $\mathcal{W}^{\chi}_{g,t}$ and $\mathcal{W}^{\chi}_{g,t} + 2\mathcal{W}^{\pi}_{g,t}$ share that sign. When the signs differ, the second factor has the sign of $\mathcal{W}^{\pi}_{g,t}$ if and only if $|\mathcal{W}^{\chi}_{g,t}| < 2|\mathcal{W}^{\pi}_{g,t}|$, in which case the product is negative. Summing the gain over dates and dividing by the cumulative loss at $\nu = 0$ gives the share. □

A.9. Corollary A2

COROLLARY A2 (Cohort Targeted Communication). *Consider the cost-push shock of Proposition 5, with realised paths $\{\hat{\pi}_t, \hat{\chi}_{g,t}\}_{t\geq 0}$. Suppose the central bank can address each cohort separately, choosing a vector of intensities $\boldsymbol{\nu} = (\nu_1, \ldots, \nu_G)$ with $\nu_g \leq 1$, so that $\kappa_g(\nu_g) = -(1-\nu_g)\Delta_g$. Assume $\Delta_g > 0$ for every g and that the news-adjusted credibility paths $X_{g,t} \equiv \hat{\chi}_{g,t} - \theta\rho_g\hat{\chi}_{g,t-1}$ are not proportional across cohorts. Then the credibility term of the wedge of Proposition 3 is zero for every cohort and every date if and only if $\nu_g = 1$ for every g. Cohort-targeted communication therefore cannot improve on the common intensity $\nu = 1$ of Proposition 5.*

PROOF. With cohort-specific intensities, the credibility term of cohort g is

$$\sigma \sum_{s\geq 0}\Big[(1-\rho_g)\,\kappa_g(\nu_g)\,X_{g,t+s} - \sum_{h}\omega_h(1-\rho_h)\,\kappa_h(\nu_h)\,X_{h,t+s}\Big],$$

where the realised paths $X_{g,t}$ do not depend on $\boldsymbol{\nu}$. The term vanishes for every g and t if and only if the products $(1-\rho_g)\kappa_g(\nu_g)\,S_{g,t}$, with $S_{g,t} \equiv \sum_{s\geq 0} X_{g,t+s}$, are equal across cohorts at every date. If $\kappa_g(\nu_g) = 0$ for every g the condition holds trivially. Conversely, suppose some $\kappa_h(\nu_h) \neq 0$. Equality across cohorts at every t then requires every other product to equal $(1-\rho_h)\kappa_h(\nu_h)S_{h,t}$, so every $\kappa_g(\nu_g)$ is non-zero and $S_{g,t}/S_{h,t}$ is constant over t for every pair, that is, the credibility paths are proportional across cohorts, which is excluded by assumption. Under the law of motion (A15) the shape of $X_{g,t}$ depends on ρ_g through $\hat{\pi}_{t-1} - \rho_g\hat{\pi}_{t-2}$ and

on $\theta\rho_g$, so the paths are proportional only when ρ_g is common across cohorts. Hence $\kappa_g(\nu_g) = 0$ for every g, and since $\Delta_g > 0$ this means $\nu_g = 1$ for every g, which is feasible under $\nu_g \le 1$. This is the common intensity of Proposition 5, so the targeted profile attains the same zero credibility loss and no targeted profile can do better. □

A.10. Proposition A1

PROPOSITION A1 (Consumption Wedges against the FIRE Path in RANK). *Consider the shock sequence of Proposition 3 and let* $\{\hat{\imath}^F_t, \hat{\pi}^F_t, dc^F_t\}_{t\ge 0}$ *be the equilibrium paths of the same RANK economy,* (23)–(25)*, under full-information rational expectations (FIRE): every cohort forecasts* $\mathbb{E}_{g,t}[\hat{\pi}_{t+1}] = \hat{\pi}^F_{t+1}$ *along the perfect-foresight path, so that the perceived and the realised real rate coincide,* $\hat{r}^F_t \equiv \hat{\imath}^F_t - \hat{\pi}^F_{t+1}$*, and every cohort chooses the same path,* $dc^F_{g,t} = dc^F_t = -\sigma \sum_{s\ge 0} \hat{r}^F_{t+s}$*. Define the* ***FIRE-referenced cohort-g consumption wedge*** *as the gap between the path cohort g chooses under its own diagnostic belief and the aggregate path of the undistorted economy,* $\mathcal{W}^F_{g,t} \equiv dc_{g,t} - dc^F_t$*. Then:*

a. Decomposition. *The FIRE-referenced wedge is the demeaned wedge of Proposition 3 plus a* ***common distortion*** Δ^F_t *that is the same for every cohort,*

$$\mathcal{W}^F_{g,t} \;\equiv\; dc_{g,t} - dc^F_t \;=\; \mathcal{W}_{g,t} + \Delta^F_t, \qquad \Delta^F_t \;\equiv\; \overline{dc}_t - dc^F_t. \tag{A29}$$

with the closed form

$$\Delta^F_t \;=\; -\sigma \sum_{s\ge 0} \big(\bar{r}^{per}_{t+s} - \hat{r}^F_{t+s}\big) \;=\; -\sigma \sum_{s\ge 0} \Big[\phi\big(\hat{\pi}_{t+s} - \hat{\pi}^F_{t+s}\big) - \big(\bar{\Lambda}^{\pi}\hat{\pi}_{t+s} + \overline{\Lambda^{\chi}\hat{\chi}}_{t+s} - \hat{\pi}^F_{t+s+1}\big)\Big]. \tag{A30}$$

where $\bar{r}^{per}_t = \hat{\imath}_t - \bar{\Lambda}^{\pi}\hat{\pi}_t - \overline{\Lambda^{\chi}\hat{\chi}}_t$ *is the population-average perceived real rate of* (A27)*. In particular* $\sum_g \omega_g \mathcal{W}^F_{g,t} = \Delta^F_t$*.*

b. Loss. *With the cohort loss* $L^F_{g,t} \equiv \frac{1}{2}(\mathcal{W}^F_{g,t})^2$*, the aggregate loss splits exactly into the cross-cohort dispersion of Proposition 3 and the common distortion,*

$$\mathcal{L}^F_t \;\equiv\; \sum_g \omega_g \tfrac{1}{2}\big(\mathcal{W}^F_{g,t}\big)^2 \;=\; \tfrac{1}{2}\,\mathrm{Var}_{\omega}\big(dc_{g,t}\big) + \tfrac{1}{2}\big(\Delta^F_t\big)^2 \;=\; \mathcal{L}_t + \tfrac{1}{2}\big(\Delta^F_t\big)^2. \tag{A31}$$

and, cumulated over the episode,

$$\|\mathcal{W}^F\|^2_{\omega} \;\equiv\; 2\sum_{t\ge 0} \mathcal{L}^F_t \;=\; \|\mathcal{W}\|^2_{\omega} + \sum_{t\ge 0}\big(\Delta^F_t\big)^2. \tag{A32}$$

c. Invariance of the reference. *The FIRE path does not depend on* $(\theta, \rho_g, \bar{\pi}_g, b, \rho_U, \nu)$. *Hence for any belief parameter* ϑ *in that list,* $\partial_\vartheta \mathcal{W}^F_{g,t} = \partial_\vartheta \mathcal{W}_{g,t} + \partial_\vartheta \Delta^F_t$ *and* $\partial_\vartheta \mathcal{L}^F_t = \partial_\vartheta \mathcal{L}_t + \Delta^F_t \, \partial_\vartheta \Delta^F_t$.

PROOF. *FIRE path.* Under FIRE the cohort Euler equation of Lemma 2 is $dc^F_{g,t} = dc^F_{g,t+1} - \sigma \hat{r}^F_t$ with $\hat{r}^F_t = \hat{\imath}^F_t - \hat{\pi}^F_{t+1}$, which contains no cohort-specific object; iterating forward with $\lim_{s\to\infty} dc^F_{g,t+s} = 0$ gives $dc^F_{g,t} = -\sigma \sum_{s\geq 0} \hat{r}^F_{t+s}$ for every g, so the cohort and the aggregate FIRE paths coincide, $dc^F_{g,t} = dc^F_t$. The FIRE paths $\{\hat{\imath}^F, \hat{\pi}^F, dc^F\}$ solve the Phillips curve, the Taylor rule $\hat{\imath}^F_t = \phi \hat{\pi}^F_t + \varepsilon_t$ and this Euler equation, a system in which none of $(\theta, \rho_g, \bar{\pi}_g, b, \rho_U, \nu)$ appears; this is part 3.

Part 1. Adding and subtracting the equilibrium aggregate path $\overline{dc}_t = \sum_g \omega_g dc_{g,t}$,

$$\mathcal{W}^F_{g,t} = \left(dc_{g,t} - \overline{dc}_t\right) + \left(\overline{dc}_t - dc^F_t\right) = \mathcal{W}_{g,t} + \Delta^F_t,$$

which is (A29); Δ^F_t carries no g index, and $\sum_g \omega_g \mathcal{W}_{g,t} = 0$ by Proposition 3 gives $\sum_g \omega_g \mathcal{W}^F_{g,t} = \Delta^F_t$. For the closed form, subtract the FIRE Euler equation from the aggregate identity (A27),

$$\Delta^F_t = -\sigma \sum_{s\geq 0} \left(\bar{r}^{\text{per}}_{t+s} - \hat{r}^F_{t+s}\right),$$

and expand both rates. The Taylor rule holds in both economies with the same monetary innovation, so $\hat{\imath}_{t+s} - \hat{\imath}^F_{t+s} = \phi(\hat{\pi}_{t+s} - \hat{\pi}^F_{t+s})$, and $\bar{r}^{\text{per}}_{t+s} - \hat{r}^F_{t+s} = \phi(\hat{\pi}_{t+s} - \hat{\pi}^F_{t+s}) - (\bar{\Lambda}^\pi \hat{\pi}_{t+s} + \overline{\Lambda^\chi \hat{\chi}}_{t+s} - \hat{\pi}^F_{t+s+1})$, which is (A30). The first bracket is the equilibrium-path gap through the policy rate, the second is the gap between the population-average forecast on the DE path and the perfect-foresight forecast on the FIRE path.

Part 2. Squaring (A29) and averaging with the population weights,

$$\sum_g \omega_g \tfrac{1}{2}\left(\mathcal{W}_{g,t} + \Delta^F_t\right)^2 = \tfrac{1}{2} \sum_g \omega_g \mathcal{W}^2_{g,t} + \Delta^F_t \sum_g \omega_g \mathcal{W}_{g,t} + \tfrac{1}{2}\left(\Delta^F_t\right)^2,$$

and the cross term vanishes because $\sum_g \omega_g \mathcal{W}_{g,t} = 0$. The first term is $\frac{1}{2}\text{Var}_\omega(dc_{g,t}) = \mathcal{L}_t$ by (47), which gives (A31); summing over t and multiplying by two gives (A32).

Part 3. Since the FIRE path is invariant to ϑ, $\partial_\vartheta dc^F_t = 0$, and differentiating (A29) and (A31) term by term gives the two identities. □

A.11. Corollary A3

COROLLARY A3 (Target Communication under the FIRE Reference). *Let* $\mathcal{L}(\nu) \equiv \sum_{t\geq 0}\mathcal{L}_t(\nu)$ *and* $\mathcal{L}^F(\nu) \equiv \sum_{t\geq 0}\mathcal{L}^F_t(\nu)$ *be the cumulative losses on the equilibrium paths of the cost-push shock at the common intensity* ν*, and* $\Delta^F_t(\nu)$ *the common distortion on those paths. Then*

$$\mathcal{L}^F(\nu) = \mathcal{L}(\nu) + \tfrac{1}{2}\sum_{t\geq 0}\big(\Delta^F_t(\nu)\big)^2, \qquad \mathcal{L}^F(0) - \mathcal{L}^F(1) = \big[\mathcal{L}(0) - \mathcal{L}(1)\big] + \tfrac{1}{2}\sum_{t\geq 0}\Big[\big(\Delta^F_t(0)\big)^2 - \big(\Delta^F_t(1)\big)^2\Big].$$

The gain from full target communication under the FIRE reference is the dispersion gain of Corollary A1 plus the change in the common distortion, which is not sign-restricted: at $\nu = 1$ *the credibility operator is zero,* $\Lambda^{\chi}_g = 0$*, but the diagnostic operator* Λ^{π}_g *still differs from the perfect-foresight forecast, so* $\Delta^F_t(1) \neq 0$ *in general. If* $\mathcal{L}$ *is minimised at* $\nu = 1$*, as in Proposition 5, then* $\nu = 1$ *minimises* $\mathcal{L}^F$ *only if*

$$\sum_{t\geq 0}\Delta^F_t(1)\,\partial_\nu\Delta^F_t(1) \;\leq\; 0,$$

and otherwise the FIRE-referenced optimum is interior, $\nu^F < 1$*: full target communication removes the dispersion but raises the common distortion at the margin.*

PROOF. The first identity is (A31) summed over t, and the second is its difference between $\nu = 0$ and $\nu = 1$. By part 3 of Proposition A1 with $\vartheta = \nu$, $\partial_\nu\mathcal{L}^F(\nu) = \partial_\nu\mathcal{L}(\nu) + \sum_t \Delta^F_t(\nu)\partial_\nu\Delta^F_t(\nu)$. If $\partial_\nu\mathcal{L}(1) = 0$, the derivative of $\mathcal{L}^F$ at $\nu = 1$ is $\sum_t \Delta^F_t(1)\partial_\nu\Delta^F_t(1)$; when this is positive, $\mathcal{L}^F(1-\epsilon) < \mathcal{L}^F(1)$ for small $\epsilon > 0$, so $\nu = 1$ is not the minimiser. That $\Delta^F_t(1) \neq 0$ in general follows from (A30) at $\nu = 1$: the credibility term drops out, but $\bar{\Lambda}^{\pi}\hat{\pi}_t = \sum_g \omega_g\rho_g[(1+\theta)\hat{\pi}_t - \theta\rho_g\hat{\pi}_{t-1}]$ is not the perfect-foresight forecast $\hat{\pi}_{t+1}$, and the DE inflation path itself differs from the FIRE path. □

REMARK A2 (Reading of the FIRE-referenced loss). *The demeaned loss of Propositions 3 and 4 is a second-order expansion of cohort utility around a budget-feasible perturbation, the equilibrium aggregate path, and it is silent about any distortion common to all cohorts. The FIRE-referenced loss replaces the reference by the allocation of the undistorted economy, so it counts both the dispersion across cohorts and the common distortion* Δ^F_t*; by* (A31) *the former measure is exactly the dispersion component of the latter. Two caveats fix its reading. First, the FIRE path is an equilibrium with a different income path, hence not a budget-feasible perturbation for a cohort of the DE economy;* $L^F_{g,t}$ *is the quadratic distance of the cohort's chosen path to the undistorted*

allocation rather than an exact second-order utility loss. Second, the FIRE path is not the first best: under a cost-push shock it fluctuates, so $\mathcal{L}^F$ measures the loss attributable to distorted beliefs, not the total welfare cost of the shock. The common distortion has two sources, visible in (A30) *and* (A34)*: the gap between the population-average forecast and the perfect-foresight forecast on a given path (the belief gap), and the difference between the DE and the FIRE equilibrium paths that the distorted forecasts induce (the equilibrium-path gap).*

A.12. Lemma ??

LEMMA **??** (The Distorted-Beliefs Jacobian). *Let $\mathcal{J}_g^{C,\pi}$ and $\mathcal{J}_g^{C,\chi}$ be the cohort-g consumption FIRE Jacobians with respect to the expected inflation path and the credibility path, and let $\widetilde{\mathcal{J}}_g^{C,\pi}$ and $\widetilde{\mathcal{J}}_g^{C,\chi}$ be the behavioural (distorted) Jacobians* (**??**)*. Let $\Lambda_g^{\pi} = \rho_g[(1+\theta)\mathbb{I} - \theta\rho_g L]$ and $\Lambda_g^{\chi} = (1-\rho_g)\kappa_g[\mathbb{I} - \theta\rho_g L]$ be the operators of* (43) *stacked over horizons, with L the lag operator. Under Assumption 2 and Proposition 2:*

a. *the cohort-g belief is $\mathrm{d}\mathbb{E}_g^{\theta}[\boldsymbol{\pi}] = \Lambda_g^{\pi}\,\mathrm{d}\boldsymbol{\pi} + \Lambda_g^{\chi}\,\mathrm{d}\boldsymbol{\chi}_g$ at every date t, with $\Lambda_{g,t}^{\pi} = \Lambda_g^{\pi}$ and $\Lambda_{g,t}^{\chi} = \Lambda_g^{\chi}$ independent of t;*
b. *every forecast-revision term in* (**??**) *is zero, $\Lambda_{g,t}^{X} - \Lambda_{g,t-1}^{X} = 0$ for all $t \geq 1$ and $X \in \{\pi, \chi\}$, and the behavioural Jacobians collapse to*

$$\widetilde{\mathcal{J}}_g^{C,\pi} = \mathcal{J}_g^{C,\pi}\Lambda_g^{\pi}, \qquad \widetilde{\mathcal{J}}_g^{C,\chi} = \mathcal{J}_g^{C,\pi}\Lambda_g^{\chi};$$

c. *the two paths reach consumption through the same perceived-rate Jacobian, so the FIRE Jacobians coincide,*

$$\mathcal{J}_g^{C,\pi} = \mathcal{J}_g^{C,\chi} = -\mathcal{J}_g^{C,r^{\mathrm{per}}} = -\mathcal{J}_g^{C,i},$$

and Λ_g^{π} and Λ_g^{χ} are the two partial derivatives of the same subjective forecast.

PROOF. a. By Proposition 2, equation (14), the linearised diagnostic expectation is

$$\mathrm{d}\mathbb{E}_{g,t+h}^{\theta}[\pi_{t+h+1}] = \rho_g\big[(1+\theta)\mathrm{d}\pi_{t+h} - \theta\rho_g\,\mathrm{d}\pi_{t+h-1}\big] + (1-\rho_g)\kappa_g\big[\mathrm{d}\chi_{g,t+h} - \theta\rho_g\,\mathrm{d}\chi_{g,t+h-1}\big] \qquad h \geq 0,$$

with ρ_g fixed by Assumption 2, θ common and $\kappa_g = -(1-\nu)\Delta_g$ fixed for a given intensity. Stacking over horizons gives $\mathrm{d}\mathbb{E}_g^{\theta}[\boldsymbol{\pi}] = \Lambda_g^{\pi}\mathrm{d}\boldsymbol{\pi} + \Lambda_g^{\chi}\mathrm{d}\boldsymbol{\chi}_g$ with the two operators of the statement, whose entries do not depend on the date at which

the expectation is formed.

b. Since $\Lambda^X_{g,t} - \Lambda^X_{g,t-1} = 0$ for $X \in \{\pi, \chi\}$, the revision term in Equation **??**, $\sum_{t\geq 1} \mathcal{R}^{C,X}_{g,t}(\Lambda^X_{g,t} - \Lambda^X_{g,t-1})$, is zero and only $\mathcal{J}^{C,X}_g \Lambda^X_{g,0} = \mathcal{J}^{C,X}_g \Lambda^X_g$ survives. Together with item 3, $\mathcal{J}^{C,\chi}_g = \mathcal{J}^{C,\pi}_g$, this proves the two collapses: the mapping from realised paths to expectations is never revised because the parameters that define it, ρ_g, θ and κ_g, are time-invariant.

c. By (20) the two paths enter the household problem only through the perceived rate, $\mathrm{d}\mathbf{r}^{\mathrm{per}}_g = \mathrm{d}\mathbf{i} - \mathrm{d}\mathbb{E}^{\theta}_g[\boldsymbol{\pi}]$, with $\partial \mathbf{r}^{\mathrm{per}}_g / \partial\, \mathbb{E}^{\theta}_g[\boldsymbol{\pi}] = -\mathbb{I}$. Applying the chain rule, the Jacobian with respect to the expected inflation path is $-\mathcal{J}^{C,r^{\mathrm{per}}}_g$ whether the expectation moves because of $\boldsymbol{\pi}$ or because of $\boldsymbol{\chi}_g$, so $\mathcal{J}^{C,\pi}_g = \mathcal{J}^{C,\chi}_g = -\mathcal{J}^{C,r^{\mathrm{per}}}_g$. Setting $\Lambda^{\pi}_g = \Lambda^{\chi}_g = 0$ leaves $\mathrm{d}\mathbf{r}^{\mathrm{per}}_g = \mathrm{d}\mathbf{i}$, so $\mathcal{J}^{C,i}_g = \mathcal{J}^{C,r^{\mathrm{per}}}_g$ and the last equality follows.

□

A.13. Proposition ??

PROPOSITION **??** (Aggregate Consumption in HANK). *Let* $\mathrm{d}\mathbf{C}_g$ *be the cohort-g consumption path generated by the Distorted-Beliefs Jacobians of Lemma* **??** *and* $\mathrm{d}\mathbf{C} = \sum_g \omega_g \mathrm{d}\mathbf{C}_g$. *Define the cohort's subjective-forecast path* $\mathrm{d}\mathcal{E}_g \equiv \Lambda^{\pi}_g \mathrm{d}\boldsymbol{\pi} + \Lambda^{\chi}_g \mathrm{d}\boldsymbol{\chi}_g$, *whose date-t element* $\mathrm{d}\mathcal{E}_{g,t} = \mathrm{d}\mathbb{E}^{\theta}_{g,t}[\pi_{t+1}]$ *is the change of cohort g's subjective inflation forecast,* $\mathrm{d}C^X_{g,t} \equiv \left[\mathcal{J}^{C,X}_g\, \mathrm{d}\mathbf{X}\right]_t$ *for* $X \in \{r, i, Y, \tau\}$, *with* $\overline{\mathrm{d}C^X}_t$ *its* ω*-weighted mean, and the two covariances*

$$\mathrm{CE}_t \equiv \mathrm{Cov}_{\omega}\big(\mathrm{MPC}_g, \mathrm{d}\mathcal{E}_{g,t}\big), \qquad \mathrm{CU} \equiv \mathrm{Cov}_{\omega}(\mathrm{MPC}_g, \mathrm{URE}_g).$$

a. *The aggregate response admits the exact decomposition*

$$\begin{aligned} \mathrm{d}C_t = & \underbrace{\left[\overline{\mathcal{J}^{C,\pi}}\,\overline{\mathrm{d}\mathcal{E}}\right]_t + \overline{\mathrm{d}C^r}_t}_{\textit{Representative Agent}} \\ & + \underbrace{\Big[\sum_g \omega_g \big(\mathcal{J}^{C,\pi}_g - \overline{\mathcal{J}^{C,\pi}}\big)\big(\mathrm{d}\mathcal{E}_g - \overline{\mathrm{d}\mathcal{E}}\big)\Big]_t}_{\textit{Cross-cohort Heterogeneity}} \\ & + \underbrace{\sum_{X\in\{i,Y,\tau\}} \overline{\mathrm{d}C^X}_t}_{\{i,Y,\tau\}\ \textit{Channel}}. \end{aligned} \tag{??}$$

b. *Because the inflation path is common across cohorts,* $\mathrm{CE}_t = \mathrm{CM}_t + \mathrm{CC}_t$*, with the diagnostic component* $\mathrm{CM}_t \equiv \mathrm{Cov}_\omega(\mathrm{MPC}_g, \Lambda^\pi_g \mathrm{d}\pi_t) = (1+\theta)\,\mathrm{Cov}_\omega(\mathrm{MPC}_g, \rho_g)\,\mathrm{d}\pi_t - \theta\,\mathrm{Cov}_\omega(\mathrm{MPC}_g, \rho_g^2)\,\mathrm{d}\pi_{t-1}$ *and the credibility component* $\mathrm{CC}_t \equiv \mathrm{Cov}_\omega(\mathrm{MPC}_g, \Lambda^\chi_g \mathrm{d}\chi_{g,t}) = -(1-\nu)\,\mathrm{Cov}_\omega\big(\mathrm{MPC}_g, (1-\rho_g)\Delta_g X_{g,t}\big)$*, with* $X_{g,t} \equiv \mathrm{d}\chi_{g,t} - \theta\rho_g \mathrm{d}\chi_{g,t-1}$*. Only* CC_t *depends on the communication intensity* ν*.*

c. *On impact, the forecast path carries no income effect, while the realised-rate channel satisfies exactly* $\sum_g \omega_g \mathcal{J}^{C,r}_{g,00} = \overline{\mathrm{MPC}}\,\overline{\mathrm{URE}} + \mathrm{CU}$*. If the cohort forecast Jacobians scale with the MPC profile, the cross-cohort heterogeneity block of* (**??**) *collapses into* $\frac{1}{\overline{\mathrm{MPC}}}\big[\overline{\mathcal{J}^{C,\pi}}\,\mathbf{CE}\big]_t$*, where* $\mathbf{CE} \equiv (\mathrm{CE}_s)_{s\geq 0}$*.*

Hence $(\mathrm{CE}_t, \mathrm{CU})$ *are sufficient statistics for determining the cohort heterogeneity on the aggregate consumption path.*

PROOF. By Lemma **??** and (36), the cohort-g response is $\mathrm{d}\mathbf{C}_g = \mathcal{J}^{C,\pi}_g \mathrm{d}\mathcal{E}_g + \mathcal{J}^{C,r}_g \mathrm{d}\mathbf{r} + \sum_{X\in\{i,Y,\tau\}} \mathcal{J}^{C,X}_g \mathrm{d}\mathbf{X}$, with $\mathrm{d}\mathcal{E}_g = \Lambda^\pi_g \mathrm{d}\boldsymbol{\pi} + \Lambda^\chi_g \mathrm{d}\boldsymbol{\chi}_g$ and the Fisher term collected as the i element of the last sum. Aggregating with the weights ω_g and applying the covariance identity to the forecast term,

$$\sum_g \omega_g \mathcal{J}^{C,\pi}_g \mathrm{d}\mathcal{E}_g = \overline{\mathcal{J}^{C,\pi}}\,\overline{\mathrm{d}\mathcal{E}} + \sum_g \omega_g \big(\mathcal{J}^{C,\pi}_g - \overline{\mathcal{J}^{C,\pi}}\big)\big(\mathrm{d}\mathcal{E}_g - \overline{\mathrm{d}\mathcal{E}}\big),$$

gives the exact decomposition (**??**).

For part 2, the date-t element of the forecast path is $\mathrm{d}\mathcal{E}_{g,t} = \rho_g(1+\theta)\mathrm{d}\pi_t - \theta\rho_g^2 \mathrm{d}\pi_{t-1} + (1-\rho_g)\kappa_g X_{g,t}$ by (43). The inflation path is common across cohorts, so the covariance with the MPC passes through the scalars $\rho_g(1+\theta)$ and $\theta\rho_g^2$, which gives CM_t; substituting $\kappa_g = -(1-\nu)\Delta_g$ gives CC_t, and ν appears nowhere else.

For part 3, the forecast enters the household problem only through the perceived rate in the Euler equation (29), never through cash-on-hand, so the forecast Jacobian carries no income effect on impact. The realised rate instead enters the budget constraint on predetermined positions, and the argument of Auclert (2019) delivers $\mathcal{J}^{C,r}_{g,00} = \mathrm{MPC}_g\,\mathrm{URE}_g$; averaging,

$$\sum_g \omega_g \mathcal{J}^{C,r}_{g,00} = \overline{\mathrm{MPC}\cdot\mathrm{URE}} = \overline{\mathrm{MPC}}\,\overline{\mathrm{URE}} + \mathrm{CU}.$$

If the cohort Jacobians scale with the MPC profile, $\mathcal{J}^{C,\pi}_g \approx (\mathrm{MPC}_g/\overline{\mathrm{MPC}})\,\overline{\mathcal{J}^{C,\pi}}$, then $\sum_g \omega_g(\mathcal{J}^{C,\pi}_g - \overline{\mathcal{J}^{C,\pi}})(\mathrm{d}\mathcal{E}_g - \overline{\mathrm{d}\mathcal{E}}) \approx \frac{1}{\overline{\mathrm{MPC}}}\,\overline{\mathcal{J}^{C,\pi}} \sum_g \omega_g(\mathrm{MPC}_g - \overline{\mathrm{MPC}})(\mathrm{d}\mathcal{E}_g - \overline{\mathrm{d}\mathcal{E}}) = \frac{1}{\overline{\mathrm{MPC}}}\,\overline{\mathcal{J}^{C,\pi}}\,\mathbf{CE}$, whose date-$t$ element is the stated collapse, which proves the sufficient-

statistic claim. By contrast, CU does not admit a closed form in the model parameters. □

A.14. Proposition 4

PROPOSITION 4 (Consumption Wedges and Aggregate Welfare in HANK). *Let* $\mathrm{d}\mathbf{C}_g$ *be the cohort-g consumption path generated by* (36)*, with the perceived-rate path in* (37)*. Define the cohort-g per-capita consumption path as* $\mathrm{d}c_{g,t} \equiv \mathrm{d}C_{g,t}/C_g$*, the Jacobians* $\mathcal{J}_g^{c,X} \equiv \mathcal{J}_g^{C,X}/C_g$*, and* $\mathcal{J}_g^{c,\pi} \equiv -\mathcal{J}_g^{c,r^{\mathrm{per}}}$*, the consumption response to the cohort's subjective inflation forecast* $\mathrm{d}\mathcal{E}_g \equiv \Lambda_g^{\pi}\mathrm{d}\boldsymbol{\pi} + \Lambda_g^{\chi}\mathrm{d}\boldsymbol{\chi}_g$*, with* Λ_g^{π} *and* Λ_g^{χ} *the stacked operators of* (43)*. Let* $\mathrm{d}\mathbf{c}_g^R$ *be the* ***rational path*** *of cohort g: its per-capita response in the same economy when it forecasts the realised inflation path with perfect foresight,* $\mathrm{d}\mathcal{E}_g^R = \mathbb{F}\,\mathrm{d}\boldsymbol{\pi}$ *with* $(\mathbb{F}\mathbf{x})_t \equiv x_{t+1}$ *the lead operator, at the same realised paths* $(\mathrm{d}\boldsymbol{\pi}, \mathrm{d}\mathbf{r}, \mathrm{d}\mathbf{X})$*,* $X \in \{i, Y, \tau\}$*. Then, the cohort consumption wedge* $\mathcal{W}_{g,t}^{H,R} \equiv \mathrm{d}c_{g,t} - \mathrm{d}c_{g,t}^R$ *is the cohort's cumulated forecast error passed through its own consumption response and splits exactly into a* ***diagnostic error*** *and a* ***credibility error****:*

$$\begin{aligned}\mathcal{W}_{g,t}^{H,R} &= \left[\mathcal{J}_g^{c,\pi}\big(\mathrm{d}\mathcal{E}_g - \mathbb{F}\,\mathrm{d}\boldsymbol{\pi}\big)\right]_t \\ &= \underbrace{\left[\mathcal{J}_g^{c,\pi}\big(\Lambda_g^{\pi} - \mathbb{F}\big)\mathrm{d}\boldsymbol{\pi}\right]_t}_{\textit{Diagnostic error}} + \underbrace{\left[\mathcal{J}_g^{c,\pi}\Lambda_g^{\chi}\,\mathrm{d}\boldsymbol{\chi}_g\right]_t}_{\textit{Credibility error}}.\end{aligned} \tag{48}$$

the balance-sheet (URE) and $\{i, Y, \tau\}$ *channels cancel because both paths face the same realised prices and income. Moreover, with the cohort and the aggregate welfare losses of* (42) *evaluated at* $\mathcal{W}_{g,t}^{H,R}$*, the aggregate loss is*

$$\mathcal{L}_t^R = \tfrac{1}{2}\,\mathrm{Var}_{\omega}\big(\mathcal{W}_{g,t}^{H,R}\big) + \tfrac{1}{2}\big(\Delta_t^{H,R}\big)^2, \qquad \Delta_t^{H,R} \equiv \sum_g \omega_g \mathcal{W}_{g,t}^{H,R} = \overline{\mathrm{d}c}_t - \overline{\mathrm{d}c^R}_t,$$

the dispersion of the cohort forecast errors plus the squared consensus error.

PROOF. The first step is to write the cohort-g consumption path. By (36), $\mathrm{d}\mathbf{C}_g = \mathcal{J}_g^{C,r}\mathrm{d}\mathbf{r} + \mathcal{J}_g^{C,r^{\mathrm{per}}}\mathrm{d}\mathbf{r}_g^{\mathrm{per}} + \mathcal{J}_g^{C,Y}\mathrm{d}\mathbf{Y}_g + \mathcal{J}_g^{C,\tau}\mathrm{d}\boldsymbol{\tau}$. By (37), the perceived-rate term has a Fisher part, $\mathcal{J}_g^{C,r^{\mathrm{per}}}\mathrm{d}\mathbf{i}$, and two belief parts, $-\mathcal{J}_g^{C,r^{\mathrm{per}}}\Lambda_g^{\pi}\mathrm{d}\boldsymbol{\pi} - \mathcal{J}_g^{C,r^{\mathrm{per}}}\Lambda_g^{\chi}\mathrm{d}\boldsymbol{\chi}_g$, with $\Lambda_g^{\pi} = \rho_g[(1+\theta)\mathbb{I} - \theta\rho_g L]$ and $\Lambda_g^{\chi} = (1-\rho_g)\kappa_g[\mathbb{I} - \theta\rho_g L]$ the stacked versions of (43). Dividing by C_g, writing $\mathcal{J}_g^{c,X} \equiv \mathcal{J}_g^{C,X}/C_g$ and $\mathcal{J}_g^{c,\pi} \equiv -\mathcal{J}_g^{c,r^{\mathrm{per}}}$, and putting the Fisher part inside

the $\{i, Y, \tau\}$ sum, the per-capita path is

$$\mathrm{d}\mathbf{c}_g = \mathcal{J}_g^{c,\pi}\big(\Lambda_g^{\pi}\,\mathrm{d}\boldsymbol{\pi} + \Lambda_g^{\chi}\,\mathrm{d}\boldsymbol{\chi}_g\big) + \mathcal{J}_g^{c,r}\,\mathrm{d}\mathbf{r} + \sum_{X\in\{i,Y,\tau\}} \mathcal{J}_g^{c,X}\,\mathrm{d}\mathbf{X}.$$

The next step is the rational path of cohort g. It is the same path with one change: the subjective forecast $\mathrm{d}\mathcal{E}_g$ is replaced by the realised next-period inflation, $\mathbb{F}\mathrm{d}\boldsymbol{\pi}$. The realised real rate, the nominal rate, income and taxes are the same paths. The Jacobians are also the same, because they are evaluated at the belief-free steady state of Lemma **??**, where the perceived and the realised rate coincide. Therefore

$$\mathrm{d}\mathbf{c}_g^{R} = \mathcal{J}_g^{c,\pi}\,\mathbb{F}\,\mathrm{d}\boldsymbol{\pi} + \mathcal{J}_g^{c,r}\,\mathrm{d}\mathbf{r} + \sum_{X\in\{i,Y,\tau\}} \mathcal{J}_g^{c,X}\,\mathrm{d}\mathbf{X}.$$

Subtracting the second expression from the first, the balance-sheet term and the $\{i, Y, \tau\}$ terms cancel, and the wedge is

$$\mathcal{W}_{g,t}^{H,R} = \Big[\mathcal{J}_g^{c,\pi}\big(\mathrm{d}\mathcal{E}_g - \mathbb{F}\mathrm{d}\boldsymbol{\pi}\big)\Big]_t = \Big[\mathcal{J}_g^{c,\pi}\big(\Lambda_g^{\pi} - \mathbb{F}\big)\mathrm{d}\boldsymbol{\pi}\Big]_t + \Big[\mathcal{J}_g^{c,\pi}\Lambda_g^{\chi}\mathrm{d}\boldsymbol{\chi}_g\Big]_t,$$

which is equation (48). The date-t element of $\mathrm{d}\mathcal{E}_g - \mathbb{F}\mathrm{d}\boldsymbol{\pi}$ is $\mathrm{d}\mathbb{E}_{g,t}^{\theta}[\pi_{t+1}] - \mathrm{d}\pi_{t+1}$, the one-step-ahead forecast error of the cohort. So the wedge is the forecast error passed through the cohort's own consumption response. The split in two parts is exact because the forecast is linear in the two channels. Note that the credibility path $\mathrm{d}\boldsymbol{\chi}_g$ is cohort-specific, so the credibility error differs across cohorts even if the Jacobians are the same for all of them. The diagnostic error differs across cohorts through ρ_g and through the Jacobian. When the per-capita Jacobians are the same for all cohorts and equal to the permanent-income operator, $[\mathcal{J}_g^{c,\pi}\mathbf{x}]_t = \sigma\sum_{s\geq 0} x_{t+s}$, the wedge becomes the RANK wedge (44) and the rational path is the same for every cohort.

Finally, the welfare loss. By (42) evaluated at the HANK wedge, the cohort loss is $L_{g,t}^{R} = \frac{1}{2}(\mathcal{W}_{g,t}^{H,R})^2$; following the steps of Proposition 3 with the cohort's rational path as the expansion point, it is again the second-order utility loss relative to that path. Multiply by ω_g, sum across cohorts and write $\Delta_t^{H,R} \equiv \sum_g \omega_g \mathcal{W}_{g,t}^{H,R} = \overline{\mathrm{d}c}_t - \overline{\mathrm{d}c^{R}}_t$. Then

$$\mathcal{L}_t^{R} = \sum_g \omega_g L_{g,t}^{R} = \tfrac{1}{2}\sum_g \omega_g\big(\mathcal{W}_{g,t}^{H,R} - \Delta_t^{H,R}\big)^2 + \tfrac{1}{2}\big(\Delta_t^{H,R}\big)^2 = \tfrac{1}{2}\,\mathrm{Var}_{\omega}\big(\mathcal{W}_{g,t}^{H,R}\big) + \tfrac{1}{2}\big(\Delta_t^{H,R}\big)^2,$$

where the cross term is zero by the definition of $\Delta_t^{H,R}$. This is what we wanted to prove. Unlike the RANK case, $\mathrm{d}c_{g,t}^R$ changes across cohorts through the Jacobians, so $\mathrm{Var}_\omega(\mathcal{W}_{g,t}^{H,R})$ is the dispersion of the forecast errors through consumption and not the dispersion of the consumption paths. The two coincide when the rational path is the same for every cohort. □

A.15. Proposition A2

PROPOSITION A2 (Consumption Wedges against the FIRE Path in HANK). *Let* $\mathrm{d}\mathbf{C}_g$ *be as in Proposition 4, and let* $\mathrm{d}\mathbf{C}_g^F$ *be the cohort-g path of the same HANK economy when every cohort forecasts with perfect foresight,* $\mathrm{d}\mathcal{E}_g^F = \mathbb{F}\,\mathrm{d}\boldsymbol{\pi}^F$ *with* $(\mathbb{F}\mathbf{x})_t \equiv x_{t+1}$ *the lead operator, on the FIRE equilibrium paths* $(\mathrm{d}\boldsymbol{\pi}^F, \mathrm{d}\mathbf{r}^F, \mathrm{d}\mathbf{X}^F)$. *Because the steady state is belief-free, Lemma* **??** *applies with the same Jacobians, and in per-capita terms* $\mathrm{d}\mathbf{c}_g^F = \mathcal{J}_g^{c,\pi}\mathbb{F}\mathrm{d}\boldsymbol{\pi}^F + \mathcal{J}_g^{c,r}\mathrm{d}\mathbf{r}^F + \sum_{X\in\{i,Y,\tau\}}\mathcal{J}_g^{c,X}\mathrm{d}\mathbf{X}^F$. *Define the* ***FIRE-referenced HANK wedge*** *against the population-average FIRE path,* $\mathcal{W}_{g,t}^{H,F} \equiv \mathrm{d}c_{g,t} - \overline{\mathrm{d}c^F}_t$. *Then:*

a. Decomposition.

$$\mathcal{W}_{g,t}^{H,F} \equiv \mathrm{d}c_{g,t} - \overline{\mathrm{d}c^F}_t = \mathcal{W}_{g,t}^{H} + \Delta_t^{H,F}, \qquad \Delta_t^{H,F} \equiv \overline{\mathrm{d}c}_t - \overline{\mathrm{d}c^F}_t, \quad \overline{\mathrm{d}c^F}_t \equiv \sum_g \omega_g\,\mathrm{d}c_{g,t}^F. \tag{A33}$$

where $\mathcal{W}_{g,t}^{H}$ *is the wedge of Proposition 4 and the common distortion is*

$$\Delta_t^{H,F} = \underbrace{\left[\overline{\mathcal{J}^{c,\pi}\Lambda^\pi}\,\mathrm{d}\boldsymbol{\pi} + \overline{\mathcal{J}^{c,\pi}\Lambda^\chi}\mathrm{d}\boldsymbol{\chi} - \overline{\mathcal{J}^{c,\pi}}\,\mathbb{F}\,\mathrm{d}\boldsymbol{\pi}^F\right]_t}_{\textit{Belief gap}} + \underbrace{\left[\overline{\mathcal{J}^{c,r}}\left(\mathrm{d}\mathbf{r} - \mathrm{d}\mathbf{r}^F\right)\right]_t + \sum_{X\in\{i,Y,\tau\}}\left[\overline{\mathcal{J}^{c,X}}\left(\mathrm{d}\mathbf{X} - \mathrm{d}\mathbf{X}^F\right)\right]_t}_{\textit{Equilibrium-path gap}}, \tag{A34}$$

so that $\sum_g \omega_g \mathcal{W}_{g,t}^{H,F} = \Delta_t^{H,F}$.

b. Loss. *With* $L_{g,t}^{H,F} \equiv \frac{1}{2}(\mathcal{W}_{g,t}^{H,F})^2$,

$$\mathcal{L}_t^{H,F} \equiv \sum_g \omega_g L_{g,t}^{H,F} = \tfrac{1}{2}\,\mathrm{Var}_\omega(\mathrm{d}c_{g,t}) + \tfrac{1}{2}\left(\Delta_t^{H,F}\right)^2 = \mathcal{L}_t^{H} + \tfrac{1}{2}\left(\Delta_t^{H,F}\right)^2.$$

c. Nesting. *Unlike the RANK case, the FIRE cohort paths differ across cohorts through the Jacobians, so the reference is their population average and not a common path. When the per-capita Jacobians are cohort-homogeneous and equal to the strict-PIH operator,* $[\mathcal{J}_g^{c,\pi}\mathbf{x}]_t = \sigma\sum_{s\geq 0} x_{t+s}$ *and the wealth channels are shut,* $\Delta_t^{H,F}$ *collapses to* Δ_t^F *of Proposition A1 and* $\mathcal{W}_{g,t}^{H,F} = \mathcal{W}_{g,t}^F$.

PROOF. *Part 1.* Adding and subtracting $\overline{\mathrm{d}c}_t$, $\mathcal{W}^{H,F}_{g,t} = (\mathrm{d}c_{g,t} - \overline{\mathrm{d}c}_t) + (\overline{\mathrm{d}c}_t - \overline{\mathrm{d}c^F}_t) = \mathcal{W}^{H}_{g,t} + \Delta^{H,F}_t$, and $\sum_g \omega_g \mathcal{W}^{H}_{g,t} = 0$ by Proposition 4 gives the population sum. For the closed form, average the DE cohort paths of (36) in per-capita terms with the weights ω_g (the proof of Proposition **??**), $\overline{\mathrm{d}\mathbf{c}} = \overline{\mathcal{J}^{c,\pi}\Lambda^{\pi}}\mathrm{d}\boldsymbol{\pi} + \overline{\mathcal{J}^{c,\pi}\Lambda^{\chi}\mathrm{d}\boldsymbol{\chi}} + \overline{\mathcal{J}^{c,r}}\mathrm{d}\mathbf{r} + \sum_X \overline{\mathcal{J}^{c,X}}\mathrm{d}\mathbf{X}$, and the FIRE cohort paths with the same weights, $\overline{\mathrm{d}\mathbf{c}^F} = \overline{\mathcal{J}^{c,\pi}}\mathbb{F}\mathrm{d}\boldsymbol{\pi}^F + \overline{\mathcal{J}^{c,r}}\mathrm{d}\mathbf{r}^F + \sum_X \overline{\mathcal{J}^{c,X}}\mathrm{d}\mathbf{X}^F$; subtracting gives (A34). The Jacobians are the same in both economies because they are evaluated at the common belief-free steady state ($\pi = \chi = 0$), where the perceived and the realised rate coincide.

Part 2. As in part 2 of Proposition A1: the cross term $\Delta^{H,F}_t \sum_g \omega_g \mathcal{W}^{H}_{g,t}$ vanishes and $\frac{1}{2}\sum_g \omega_g (\mathcal{W}^{H}_{g,t})^2 = \frac{1}{2}\mathrm{Var}_{\omega}(\mathrm{d}c_{g,t}) = \mathcal{L}^{H}_t$ by Proposition 4.

Part 3. With homogeneous strict-PIH Jacobians and no wealth channels, $\mathrm{d}c^F_{g,t} = -\sigma \sum_{s\geq 0}(\mathrm{d}i^F_{t+s} - \mathrm{d}\pi^F_{t+s+1}) = dc^F_t$ for every g, so $\overline{\mathrm{d}c^F}_t = dc^F_t$, and the belief gap of (A34) becomes $\sigma \sum_{s\geq 0}(\bar{\Lambda}^{\pi}\hat{\pi}_{t+s} + \overline{\Lambda^{\chi}\hat{\chi}}_{t+s} - \hat{\pi}^F_{t+s+1})$ while the equilibrium-path gap reduces to the Fisher term $-\sigma \sum_{s\geq 0}(\hat{\imath}_{t+s} - \hat{\imath}^F_{t+s})$; their sum is (A30). □

A.16. Proposition A3

PROPOSITION A3 (Optimal Target Communication in HANK). *Consider the cost-push shock of Proposition 5, with realised paths $\{\mathrm{d}\pi_t, \mathrm{d}\chi_{g,t}\}_{t\geq 0}$, and the HANK wedge of Proposition 4. Since ν enters only $\kappa_g = -(1-\nu)\Delta_g$, the ex-post credibility contribution to the cohort-g wedge is*

$$\mathcal{W}^{H,\chi}_{g,t}(\nu) = \left[\mathcal{J}^{c,\pi}_g \Lambda^{\chi}_g \mathrm{d}\boldsymbol{\chi}_g - \overline{\mathcal{J}^{c,\pi}\Lambda^{\chi}\mathrm{d}\boldsymbol{\chi}}\right]_t = -(1-\nu)\,\mathcal{D}^{\mathcal{J}}_{g,t},$$
$$\mathcal{D}^{\mathcal{J}}_{g,t} \equiv \left[(1-\rho_g)\Delta_g\, \mathcal{J}^{c,\pi}_g \mathbf{X}_g\right]_t - \sum_h \omega_h \left[(1-\rho_h)\Delta_h\, \mathcal{J}^{c,\pi}_h \mathbf{X}_h\right]_t,$$

*where $\mathbf{X}_g \equiv (\mathbb{I} - \theta\rho_g L)\mathrm{d}\boldsymbol{\chi}_g$ and $\mathcal{D}^{\mathcal{J}}_{g,t}$ is the **MPC-scaled credibility exposure**, the Jacobian-weighted counterpart of $D_{g,t}$ in the proof of Proposition 5, which does not depend on ν. Let $\mathcal{L}^{H,\chi}_t(\nu) = \frac{1}{2}\sum_g \omega_g (\mathcal{W}^{H,\chi}_{g,t})^2$. Then:*

a. Common intensity. *The credibility loss is $\mathcal{L}^{H,\chi}_t(\nu) = (1-\nu)^2\,\frac{1}{2}\,\mathrm{Var}_{\omega}(\mathcal{D}^{\mathcal{J}}_{g,t})$, so the intensity minimising $\sum_t \mathcal{L}^{H,\chi}_t(\nu)$ is $\nu^{\star}_{\mathrm{agg}} = 1$ for any profile of cohort Jacobians. Heterogeneous MPCs change the size of the credibility loss, $\frac{1}{2}\sum_t \mathrm{Var}_{\omega}(\mathcal{D}^{\mathcal{J}}_{g,t})$, and which cohorts carry it, but not the intensity that removes it.*

b. Cohort-specific intensity. *With cohort-specific intensities $\nu_g \leq 1$ as in Corollary A2, the credibility term is zero for every cohort and date if and only if $\nu_g = 1$ for every g, unless the MPC-scaled credibility responses $[\mathcal{J}^{c,\pi}_g \mathbf{X}_g]_t$ are proportional across cohorts.*

Targeted communication cannot improve on the common optimum.
When the per-capita Jacobian is cohort-homogeneous and equal to the permanent-income response, $[\mathcal{J}_g^{c,\pi}\mathbf{X}_g]_t = \sigma\sum_{s\geq 0} X_{g,t+s}$, then $\mathcal{D}^{\mathcal{J}}_{g,t} = \sigma\sum_{s\geq 0} D_{g,t+s}$ and both parts collapse to Proposition 5 and Corollary A2.

PROOF. Following Proposition 5, fix the realised paths $d\boldsymbol{\pi}$ and $d\boldsymbol{\chi}_g$ induced by the cost-push shock. By Proposition 4, the credibility contribution to the cohort-g wedge is $[\mathcal{J}_g^{c,\pi}\Lambda_g^{\chi}d\boldsymbol{\chi}_g]_t$ demeaned across cohorts, and by (43) stacked over horizons, $\Lambda_g^{\chi}d\boldsymbol{\chi}_g = (1-\rho_g)\kappa_g\mathbf{X}_g$ with $\mathbf{X}_g = (\mathbb{I} - \theta\rho_g L)d\boldsymbol{\chi}_g$. Substituting $\kappa_g = -(1-\nu)\Delta_g$, common ν across cohorts,

$$\mathcal{J}_g^{c,\pi}\Lambda_g^{\chi}d\boldsymbol{\chi}_g = -(1-\nu)\,(1-\rho_g)\Delta_g\,\mathcal{J}_g^{c,\pi}\mathbf{X}_g,$$

and demeaning gives $\mathcal{W}^{H,\chi}_{g,t}(\nu) = -(1-\nu)\mathcal{D}^{\mathcal{J}}_{g,t}$, where the realised response $[\mathcal{J}_g^{c,\pi}\mathbf{X}_g]_t$ is independent of ν. This is the exact analogue of the scaling step in the proof of part 1 of Proposition 5, with each cohort's exposure weighted by its own per-capita Jacobian.

Taking the ω-weighted mean of the squares as in Proposition 4,

$$\mathcal{L}_t^{H,\chi}(\nu) = \tfrac{1}{2}\sum_g \omega_g\big(\mathcal{W}^{H,\chi}_{g,t}\big)^2 = (1-\nu)^2\,\tfrac{1}{2}\,\mathrm{Var}_{\omega}\big(\mathcal{D}^{\mathcal{J}}_{g,t}\big),$$

so the first-order condition of $\sum_t \mathcal{L}_t^{H,\chi}(\nu)$ with respect to ν is $-(1-\nu)\sum_t \mathrm{Var}_{\omega}(\mathcal{D}^{\mathcal{J}}_{g,t}) = 0$, which gives $\nu^{\star}_{\text{agg}} = 1$ whenever some $\mathcal{D}^{\mathcal{J}}_{g,t} \neq 0$; the second derivative is positive, so it is the unique minimiser. The Jacobians enter only through the variance, which proves part 1.

For part 2, with cohort-specific intensities the credibility term of cohort g is $[(1-\rho_g)\kappa_g(\nu_g)\,\mathcal{J}_g^{c,\pi}\mathbf{X}_g]_t$ demeaned, and the argument of Corollary A2 applies verbatim with $[\mathcal{J}_g^{c,\pi}\mathbf{X}_g]_t$ in place of $S_{g,t}$: the term vanishes for every g and t if and only if the products $(1-\rho_g)\kappa_g(\nu_g)[\mathcal{J}_g^{c,\pi}\mathbf{X}_g]_t$ are equal across cohorts at every date, which, unless the responses are proportional across cohorts, forces $\kappa_g(\nu_g) = 0$, that is, $\nu_g = 1$ for every g.

Finally, when $[\mathcal{J}_g^{c,\pi}\mathbf{X}_g]_t = \sigma\sum_{s\geq 0} X_{g,t+s}$ for every g, the exposure collapses to $\mathcal{D}^{\mathcal{J}}_{g,t} = \sigma\sum_{s\geq 0} D_{g,t+s}$ and the credibility wedge to $-(1-\nu)\sigma\sum_{s\geq 0} D_{g,t+s}$, exactly the ex-post credibility wedge of part 1 of Proposition 5; part 1 then restates its part 2 and part 2 restates Corollary A2. □

Appendix B. Equilibrium Definition

DEFINITION A1 (Competitive equilibrium). *Given the initial cohort distributions* $\{D_{g,0}\}_g$ *and exogenous paths* $\{\varepsilon_t^{MP}, \varepsilon_t^{CP}\}$*, an equilibrium is a collection of value and policy functions* $\{V_{g,t}, c_{g,t}, a_{g,t}\}_g$*, distributions* $\{D_{g,t}\}_g$*, prices* $\{w_t, r_t, i_t, \pi_t, \pi_t^w\}$*, cohort credibility sequences* $\{\chi_{g,t}\}_g$*, policies* $\{\tau_t\}$*, and aggregates* $\{C_t, N_t, A_t, Y_t\}$ *such that: (i)* $\{c_{g,t}, a_{g,t}\}$ *solve* (28) *under the diagnostic expectations of Proposition 2 and the perceived rate* (27)*; (ii)* $\{D_{g,t}\}$ *satisfy* (30)*; (iii) the firm and union optimality conditions* $w_t = \Theta_t$ *and* (32) *hold, with* $Y_t = \Theta_t N_t$ *and* $N_{g,t} = \Gamma_g(N_t)$*; (iv) the government budget and the Taylor rule* (33) *hold; (v) cohort credibility* $\chi_{g,t}$ *evolves according to the law of motion* (A15)*; and (vi) the bond and goods markets clear,*

$$A_t = \sum_g \omega_g A_{g,t} = B, \qquad Y_t = \sum_g \omega_g C_{g,t} + G. \tag{A35}$$

Appendix C. Empirical Data

This appendix shows in detail how the empirical variables mentioned in Section 2, and used throughout the paper, are constructed. It covers the survey questions and sample, the inflation history, the wealth and MPC data and some descriptive statistics.

C.1. Survey Questions

This section reproduces the questions of the Bank of England / Ipsos Inflation Attitudes Survey (IAS) that are use along the empirical analysis of Section 2. The survey is administered to a repeated cross-section of UK adults. Each response has assigned a post-stratification weight, used as the observation weight throughout. In what follows, the relevant sections of the survey for this paper are presented.

C.1.1. Coverage

The IAS provides quarterly data, collected every February, May, August and November since 2001Q1 and it is currently conducted by Ipsos. Each wave is a new, nationally representative sample of UK adults, so the data form a repeated cross-section rather than a panel. Cohorts are therefore not followed at the individual level.

Three distinct windows are used in the empirical section.

a. *Structural baseline.* All waves up to and including 2021Q4. I choose this range for two reasons. First, it leaves out the recent global inflation surge, which contaminates the inflation-experience data and artificially inflates the mean and variance of the younger cohorts. Second, it avoids the 2022 redesign of several survey variables, which would break the comparability with the historical data—for example, the change in the age and income bands and the 2022 May extension of the inflation-expectation scale to codes 20–25.
b. *Cue-mix cross-section* (Question 2aiv). Available in only six waves — 2017Q1, 2023Q1, 2024Q1, 2025Q1, 2025Q3 and 2026Q1. Provides a separate sample from the structural baseline.
c. *Full-sample robustness.* Full sample up to 2026Q1, mainly used for the post-2022 robustness checks, but not for the headline calibration.

Throughout the paper, the post-stratification weights provided by the survey are used. Thus, each cross-section aligns with the UK population profile by age, gender, working status and region. Next, I reproduce the particular questions of interest used throughout.

C.1.2. Demographics

Age `[age]`.. Six ordinal bands that index the cohort $g \in \{1, \ldots, 6\}$:

- 1 = 15–24,
- 2 = 25–34,
- 3 = 35–44,
- 4 = 45–54,
- 5 = 55–64,
- 6 = 65+.

The precessed age band fixes the cohort lifetime memory window Ω_g, with memory beginning at age 16 following Gennaioli et al. (2024). A caveat emerges since from 2022 the online omnibus uses bins for ages 16–75, recoded onto the original six bands. We keep waves ≤ 2021Q4 in the baseline so this recoding does not affect the estimation sample.

Gross Household Income `[income]`.. Annual gross household income window. The coding changes in 2016 and again in 2022 as follows:

- 1 = <£9,500 *[removed 2022 Feb]*,
- 2 = £9,500–17,499 *[removed 2022 Feb]*,
- 3 = £17,500–24,999 *[removed 2022 Feb]*,

- 4 = >£25,000 *[removed 2016 Feb]*,
- 5 = £25,000–39,999 *[added 2016 Feb, removed 2022 Feb]*,
- 6 = >£40,000 *[added 2016 Feb, removed 2022 Feb]*,
- 7–11 = post-2022 bands (<£9,999; £10,000–19,999; £20,000–34,999; £35,000–44,999; >£45,000) *[added 2022 Feb]*,
- 12 = prefer not to answer *[added 2022 Feb]*.

The pre-2022 sample collapse the six bands into three groups: LOW = $\{1, 2\}$ (<£17,500), MID = $\{3\}$ (£17,500–24,999), HIGH = $\{4, 5, 6\}$ (>£25,000).

FInally, the BoE provides post-stratification weights that aligns the sample with national population profiles by age, gender, working status, and region. Those weights are used in every regression and weighted mean.

C.2. Perceived Inflation

Question 1 — perceived inflation over the past 12 months [`q1`] .

- *"Which of these options best describes how prices have changed over the last 12 months?"*

This question captures the household's perception of realised inflation over the preceding year, with the following options to pick from:

- 1 = gone down,
- 2 = not changed,
- 3 = up by 1% or less,
- 4 = up by 1–2%,
- 5 = up by 2–3%,
- 6 = up by 3–4%,
- 7 = up by 4–5%,
- 8 = up by 5% or more,
- 9 = don't know.

I map each bind to its midpoint in percent for the codes 3–7. The answers "not changed" (code 2) is set to zero and "gone down" (code 1) to −0.5%, while the top band "up by 5% or more" (code 8) is top-coded at 6.5%. The "don't know" code (9) is set to a missing value. This yields the perceived twelve-month inflation $\pi_{i,t}^{\text{perc}}$. Finally, the coverage runs for the full 2001Q1 – 2026Q1 sample.

C.2.1. Inflation Expectations

Question 2a — one-year ahead inflation expectation `[q2a_agg1]`.

- *"How much would you expect prices in the shops generally to change over the next 12 months?".*

Question `q2` offers a scale (go down / not change / up by 1% bands up to "5% or more"), with follow-ups capturing larger rises and falls and a further >10% extension. The BoE harmonises these into the aggregate variable `q2a_agg1`, with the following options to pick from:

- 1 = Down by 1% or less,
- 2 = Down by 1–2%,
- 3 = Down by 2–3%,
- 4 = Down by 3–4%,
- 5 = Down by 4–5%,
- 6 = Down by 5% or more,
- 7 = not changed,
- 8 = up by 1% or less,
- 9 = up by 1–2%,
- 10 = up by 2–3%,
- 11 = up by 3–4%,
- 12 = up by 4–5%,
- 13 = up by 5–6%,
- 14 = up by 6–7%,
- 15 = up by 7–8%,
- 16 = up by 8–9%,
- 17 = up by 9–10%,
- 18 = up by 10% or more,
- 19 = no idea,
- 20 = up by 10–11%,
- 21 = up by 11–12%,
- 22 = up by 12–13%,
- 23 = up by 13–14%,
- 24 = up by 14–15%,
- 25 = up by 15% or more *(codes 20–25 added 2022 May)*.

To process this question, I map each band to its midpoint in percent per year and set the "no idea" code (19) to a missing value. This yields the one-year expectation $\mathbb{E}_g[\pi_{t+1}]$. Because the baseline sample is ≤ 2021Q4, codes 20–25 do not show up in

the sample and the relevant range is codes 1–18 plus the dropped code 19.

Question 2c — five-years ahead inflation expectation [q2c_agg1] .

- *"And how about the longer term, say in five years time? How much would you expect prices in the shops generally to change over a year then?"*

Recorded and following the same structure as in `q2a_agg1` above. Similarly, to process it a midpoint map is used and "no idea" is set to missing value. This variable is labeled as $\mathbb{E}_g[\pi_{t+20}]$.

Question 2aiii — recent revision of inflation expectations [q2aiii, q2ciii]. .

- *"How has your expectation [of expected price change over the next 12 months] changed over the past 12 months?"*

Asked only to those who gave a numeric answer (codes 1–8) to Question 2a(1y)/2c(5y). The options are:

- 1 = Risen a lot,
- 2 = Risen a little,
- 3 = Unchanged,
- 4 = Fallen a little,
- 5 = Fallen a lot,
- 6 = No idea.

Collapsed to an up (if 1 or 2 was chosen)/down(if 4 or 5 was chosen) indicator.

Question 2aiv — self-reported forecast cues [q2aiv1–q2aiv13].

- *"What were the most important factors that led you [to revise / not revise] your expectation of prices in the shops over the next 12 months? Please select up to 4."*

First asked 2017 Feb to those who answered 1–5 to in Question 2aiii. The response is stored as thirteen binary indicators $d_{i,t}^{(k)} \in \{0, 1\}$:

a. How prices have changed in the shops recently, over the last 12 months;
b. How prices have changed in the shops, on average, over the longer term, i.e. the last few years;

c. Reports of current inflation in the media;
d. Discussion of the prospects for inflation in the media;
e. The level of interest rates;
f. The inflation target set by the government;
g. The current strength of the UK economy;
h. Expectations about how economic conditions in the UK are likely to evolve;
i. The level of the exchange rate (the value of sterling);
j. Other factors;
k. *(unused code)*;
l. How the price of energy (e.g. gas, electricity or liquid fuel bills) has changed recently, over the last 12 months *[added 2023 Feb]*;
m. How the price of petrol or diesel has changed recently, over the last 12 months *[added 2023 Feb]*.

When using this question, it is important to explicitly say that the sample for 2aiv has only six waves: 2017Q1, 2023Q1, 2024Q1, 2025Q1, 2025Q3 and 2026Q1. The cue-mix test is therefore a separate cross-section from the ≤ 2021Q4 structural baseline. Finally, I aggregate the cues into only four groups:

- *target* (θ_T: option 6),
- *instrument* (θ_I: option 5),
- *experience* (θ_E: option 2),
- *shock* (θ_S: options 1, 12, 13)

C.3. One-Year-Forecasted Nominal Interest Rate

Question 21 — expected Bank Rate [q21_1, q21_2, q21_3].

- *The level of interest rates set by the Bank of England is currently [X]%. At what level do you expect them to be in one/two/five years' time?"*

I use the expected level of the Bank Rate over the one-year horizon `q21_1`. Answers are reported in 1–31 coded bins and uses the midpoint. This yields the expected nominal Bank Rate $\mathbb{E}_g[i_{t+1}]$.

Coverage.. Question 21 is asked in the February waves only, from 2014 to 2026.

C.3.1. Behavioural response and credibility

Question 17 — actions expected to be taken [q17_1–q17_8].

- *"Which, if any, of the following actions are you taking, or planning to take, in the light of your expectations of price changes over the next twelve months? You can choose up to three."*

This question is stored as eight binary indicators:

a. Bring forward major purchases such as furniture or electrical goods;
b. Cut back spending and save more;
c. Shop around more for better value goods and services;
d. Push for increased pay with current employer;
e. Look to increase income in other ways (e.g. change jobs, take on a second job, work more hours with current employer);
f. Move savings out of banks or building societies into other assets such as shares, bonds, housing or gold;
g. Take no action;
h. None / Don't know.

I use the three options that map to the consumption–saving margin decompose them by income and the 3 age groups:

a. `q17_1` (bring forward purchases — Euler-Motive),
b. `q17_2` (cut back spending and save more — precautionary),
c. `q17_7` (take no action — baseline case).

Question 18 — anchoring / credibility [`q18`].

- *"How confident are you that inflation will be close to the 2% target set by the government in two to three years' time? By close I mean that inflation will be between 1% and 3%."*

  Available options:

- 1 = Very confident (better than 50% chance),
- 2 = Fairly confident (around 50% chance),
- 3 = Not very confident (around 25% chance),
- 4 = Not at all confident (less than 25% chance),
- 5 = Don't know.

Note. The single 2012Q1 wave used an earlier "chance" wording, since removed, mapped onto the same four ordinal levels: 6 = very good chance (>75%); 7 = good chance (50–75%); 8 = some chance (25–50%); 9 = not much chance (<25%). To process this question, I drop "don't know" (code 5), map the remaining codes to $q18^{\text{ord}} \in \{1, \ldots, 4\}$ (with the old-wave codes 6–9 mapped to 1–4), and make a

de-anchoring indicator $u_{i,t} = \mathbf{1}\{q18^{\text{ord}} \geq 3\}$. Finally, the mean $P^{\text{unanch}}_{w,g} = \mathbb{E}_{w,g}[u_{i,t}]$ gives the credibility weight $\chi_t = 1 - P^{\text{unanch}}_t$ at the aggregate level. Within each cohort cell the mean is survey-weighted; the aggregate χ_t then combines the cohort cells using their respondent counts.

C.4. Inflation Memories

Inflation Memories are calculated using the official quarterly UK inflation rate provided by the Office for National Statistics. The constructed series spans from 1949Q2 to 2026Q1 and covers every cohort's lifetime memory window. First, I combine the monthly historical modelled index `JFDN` (*CPI HISTORICAL MODELLED INDEX 00, 1965=100*, Jan 1949 – Dec 1988) with the recent index `D7BT` (*CPI INDEX 00, 2015=100*, monthly 1988 – Mar 2026).

Following Gennaioli et al. (2024), the series is annualised quarter-on- quarter growth rate of the end-of-quarter CPI level, $\pi_t = ((\text{CPI}_t/\text{CPI}_{t-1})^4 - 1) \times 100$ (the annualized convention is used for the cue π_t that enters in the IRF's). Consequently, for every individual i in cohort g surveyed in quarter w, the per-cohort historical database is calculated as the sequence of quarterly annualised inflation observations $\Pi^{\text{hist}}_{g,w} = \{\pi_{w-2-s}\}_{s=0}^{a^Q_g - 1}$, with $a^Q_g = 4\,(a_g - 16)$ quarters of memory (memory starting at age 16 following Gennaioli et al. (2024)), and $a_g \in \{20, 30, 40, 50, 60, 70\}$ the cohort age midpoint. Finally, the current cue is $\pi_t \equiv \pi_{w-1}$, namely, the most recent completed quarter before the survey, so the database ends one quarter before the cue and never overlaps it.

C.5. Wealth and MPCs

A key component of the HANK model is the covariance $\text{Cov}(\text{MPC}_g, \lambda_g) < 0$ (Section 4.2). To construct the marginal propensity to consume (MPC), I use household wealth from the ONS *Wealth and Assets Survey* (WAS) together with the cohort income profile already provided by the BoE IAS. Both objects are defined at the cohort level $g \in \{1, \ldots, 6\}$, matching the survey age bands.

Wealth.. I take per-cohort total net household wealth from the ONS Wealth and Assets Survey, measured as the population-weighted mean within the corresponding age bands. To stay consistent with the *structural baseline*, the calibration uses the pre-pandemic round (April 2018 – March 2020), which matches the memory

database frozen at 2019Q4. The published wealth bands are then aggregated to the six IAS cohorts, delivering the wealth profile W_g.

Reduced-form MPC.. Dividing wealth by the cohort income profile Y_g from the IAS income question (Section C.1) gives the cohort wealth-to-income ratio $(W/Y)_g$, which I map to a marginal propensity to consume through the reduced form

$$\mathrm{MPC}_g \;=\; \frac{1}{1 + k\,(W/Y)_g}.$$

Thus, cohorts with little wealth relative to income behave close to hand-to-mouth ($\mathrm{MPC}_g \to 1$), while wealth-rich cohorts smooth consumption (MPC_g small). The single scale parameter k is calibrated so that the population-weighted aggregate reproduces the UK consensus value $\overline{\mathrm{MPC}} \approx 0.30$, and the resulting cohort MPCs span 0.09–0.72.

C.5.1. Descriptive Statistics

TABLE A1. **Data Sources and Descriptive statistics.**

Variable	N	Mean	SD	Min	Max
Panel A: Inflation Expectations					
BoE/NOP Inflation Attitudes Survey | individual | quarterly | 2001Q1–2021Q4 (78 waves, n = 203, 037)					
$\mathbb{E}_i[\pi_{t+4}]$ (1y point forecast, %)	113,973	3.33	2.78	-5.50	10.50
$\mathbb{E}_i[\pi_{t+20}]$ (5y point forecast, %)	86,920	3.62	2.94	-5.50	10.50
survey weight ω_i	203,037	1.00	0.4582	0.0011	13.49
Panel B: Recall Regressors (GLSS)					
ONS CPI | cohort × wave | quarterly | 2001Q1–2021Q4 (6 cohorts, 1,044 cells)					
$\bar{\pi}_g$ (cohort norm, %)	1,044	5.19	2.74	1.07	17.30
V_g (lifetime inflation variance)	1,044	27.40	19.02	1.38	103.6
$\Delta_g = \pi^{\text{cue}} - \bar{\pi}_g$ (cue gap, %)	1,044	-1.21	4.76	-13.54	37.71
$X^1_{i,t}$ (primacy regressor)	1,044	0.1411	0.6359	-2.24	2.07
$X^2_{i,t}$ (recency regressor)	1,044	0.1604	0.6009	-2.35	1.83
$X^4_{i,t}$ (cue regressor of γ_{cue})	1,044	378.1	548.9	-2,642.3	3,185.2
Panel C: Credibility					
BoE IAS Q18 | wave aggregate | annual | 2011Q1–2021Q1 (11 waves)					
$\chi_t = 1 - P_t^{\text{unanch}}$ (anchoring share)	11	0.4645	0.1046	0.2527	0.5852
$\lvert\Delta\chi_t\rvert$ (abs. wave-to-wave change)	10	0.0722	0.0423	0.0078	0.1367
Panel D: Inflation Experience Series					
ONS CPI | aggregate | quarterly-annualised | 1949Q2–2026Q1 (308 quarters)					
π_t (Q-annualised CPI inflation, %)	308	4.73	5.49	-6.95	43.10
Panel E: Cohort Wealth & Income					
ONS WAS + BoE IAS income | cohort | cross-section | Apr 2018–Mar 2020 (6 cohorts)					
W_g (mean individual wealth, £000)	6	276.1	217.4	22.45	548.1
Y_g (mean cohort income, £000)	6	23.92	3.94	17.60	27.71
$(W/I)_g$ (wealth-to-income ratio)	6	12.21	10.80	1.07	27.39
MPC_g (reduced form, $\overline{\text{MPC}} = 0.30$)	6	0.3167	0.2512	0.0918	0.7211

Note: Panel sub-headers: every variable is grouped by source | level of observation | frequency | sample period.

C.6. Calibration of the Belief Parameters

This section presents how the parameters of the belief block reported in Table 1 are obtained. The cohort moments ρ_g and $\bar{\pi}_g$ are measured from each cohort's lifetime inflation. The learning parameters ρ_U and b are imported from Hommes and Lustenhouwer (2019), and the diagnostic parameter θ from Carroll (2003). Finally, the response of credibility to inflation in the BoE Inflation Attitudes Survey is used as an untargeted moment to assess the calibration.

C.6.1. Cohort Moments, ρ_g and $\bar{\pi}_g$

Under Assumption 2, cohort g perceives inflation as a first-order autoregression with persistence ρ_g, and under Assumption 3 its experiential anchor is built on its lived inflation $\bar{\pi}_g$. Both moments are computed from the cohort's lifetime database of quarterly-annualised CPI inflation, namely the $a_g^Q = 4\,(a_g - 16)$ quarters before 2019Q4, with $a_g \in \{20, 30, 40, 50, 60, 70\}$ the cohort age midpoint. The lived inflation $\bar{\pi}_g$ is the mean of that database, and ρ_g is the OLS slope of

$$\pi_s = c_g + \rho_g\, \pi_{s-1} + e_{g,s}, \tag{A36}$$

estimated on the same a_g^Q observations. Panel B of Table 1 reports both moments. The two moments are measured once, at 2019Q4, and kept fixed over the horizon of the model.

Two limitations are worth noting. First, the window of the youngest cohort has only 16 quarters, so its persistence is imprecise: the estimates of the three youngest cohorts, between 0.04 and 0.11, are not statistically different from zero, while those of the two oldest cohorts, 0.56 and 0.55, are tightly estimated. Second, the windows are nested, since every cohort's database contains the database of the younger ones, so the six estimates are not independent.

C.6.2. Learning Parameters, ρ_U and b

Equation (8) ties b to the perceived innovation variance σ_g^2 and to ρ_U. Since σ_g^2 is a perceived variance and not an estimated one, b is not computed from it. Instead, both learning parameters are imported from Hommes and Lustenhouwer (2019), whose measure of credibility has the same logit form as Equation (8) and whose performance measure discounts past squared forecast errors at a geometric rate,

as in Equation (9):

- $\rho_U = 0.5$ is their memory parameter.
- $b = 0.25$ is their intensity of choice. They set it to 40,000 for quarterly inflation expressed in decimals. The forecast errors here are in percentage points of annualised inflation, which are 400 times larger, so the same intensity is $b = 40{,}000/400^2 = 0.25$.

C.6.3. Diagnostic Parameter, θ

The diagnostic parameter is common across cohorts and imported from Carroll (2003). By Lemma A1, the diagnostic expectation can be written as

$$\mathbb{E}^{\theta}_{g,t}[\pi_{t+1}] = \mathbb{E}_{g,t-1}[\pi_{t+1}] + (1+\theta)\big(\mathbb{E}_{g,t}[\pi_{t+1}] - \mathbb{E}_{g,t-1}[\pi_{t+1}]\big),$$

so it absorbs a fraction $1+\theta$ of the forecast revision on impact. Carroll (2003) estimates that a fraction $\lambda = 0.27$ of households update their inflation expectation each quarter, so the mean household expectation absorbs the fraction λ of the news on impact. Equating the two gives

$$\theta = \lambda - 1 = -0.73.$$

The mapping is exact on impact only. After the first quarter the diagnostic expectation absorbs the rest of the revision, while under sticky information the revision keeps being absorbed gradually.

C.6.4. Untargeted Moment: Inflation De-anchors Credibility

Since the learning parameters are imported, the response of credibility to inflation in the data is not used in the calibration and serves as a check on it. The measure of credibility is the aggregate anchoring share χ_t, the share of respondents who answer question Q18 of the BoE Inflation Attitudes Survey "very" or "fairly" confident that inflation will be close to the target (Section C). The question is asked once a year, so the sample has 17 annual waves, 2011Q1–2026Q1. I regress χ_t on its own lag and on the lagged inflation gap,

$$\chi_t = c + \rho_y \chi_{t-1} - \eta_y (\pi_{t-1} - \pi^*) + \varepsilon_t, \tag{A37}$$

where the lag is the previous annual wave and π_t is quarterly-annualised CPI

inflation in percentage points, which gives 16 observations. The estimated parameters, with Newey–West standard errors in parentheses, are $\rho_y = 0.317$ (0.220) and $\eta_y = 0.0163$ (0.0070), with $R^2 = 0.45$. The implied long-run response of the anchoring share to a permanent one-percentage-point inflation gap is

$$d\chi/d\pi|_{LR} = -\eta_y/(1-\rho_y) = -0.024.$$

A static regression of χ_t on the contemporaneous inflation gap, which estimates the same long-run response without relying on ρ_y, gives -0.030 (0.008).

The model counterpart follows from the law of motion of cohort credibility in Equation (A15). A permanent unit inflation gap is a permanent quasi-innovation of $1-\rho_g$, so the long-run response of aggregate credibility at $\nu = 0$ is

$$\left.\frac{d\chi}{d\pi}\right|_{LR} = -\frac{1}{100}\sum_g \omega_g \frac{(1-\rho_g)\,\xi_g}{1-\rho_U} = -0.039, \tag{A38}$$

where the factor 100 converts the decimal units of $\hat{\pi}_t$ into percentage points. Hence, the calibrated model de-anchors somewhat more than the point estimates, about 1.6 times the dynamic estimate and 1.3 times the static one, and its response lies inside the 95% interval of the static estimate, $[-0.045, -0.015]$. Regarding persistence, $\rho_U = 0.5$ implies an annual persistence of credibility of $0.5^4 = 0.06$, below the point estimate $\rho_y = 0.317$ but within 1.2 standard errors of it.

These comparisons should be read with caution. The regressions rest on 16 or 17 annual observations, the persistence ρ_y cannot be distinguished from zero, and most of the variation in the inflation gap comes from the 2022–2024 episode.

Appendix D. Additional Empirical Facts

This appendix documents two additional facts important for the transmission mechanism of beliefs. Fact A1 shows that when households expect more inflation they report changing the timing of their consumption, on both the spending and the saving margin. This provides evidence of an intertemporal Fisher effect in place. Fact A2 shows that the same cohorts that lived through higher inflation perceive higher inflation today and, since the 2021–22 surge, have become more sceptical that higher interest rates reduce prices.

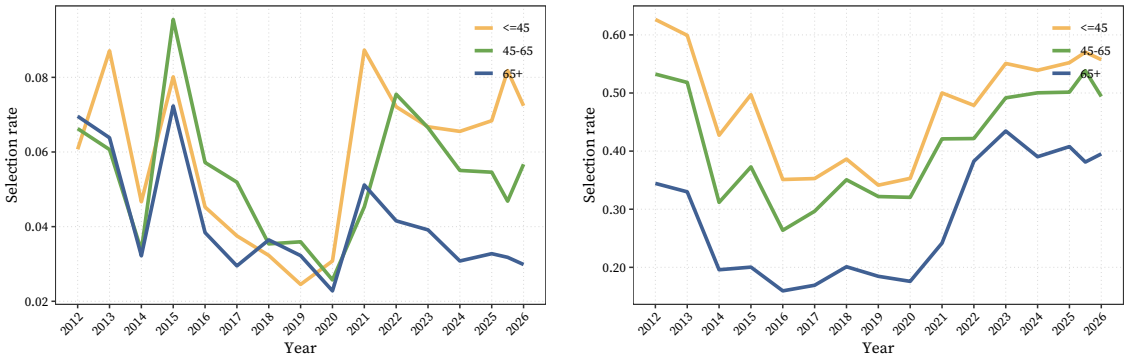
Fact A1. Planned-Consumption Pass-Through.

"When expected inflation rises, a larger share of households report higher intentions to bring forward major purchases and to cut back and save more."

The distorted beliefs channel stated in the paper is relevant only if revisions in expectations trigger changes in consumption decisions. The standard mechanism operates through the cohort Euler equation: a household that expects higher inflation perceives a lower real interest rate, which increases the incentive to bring consumption forward. Therefore, in theory, heterogeneity in expectations would directly drive heterogeneity in consumption timing. The link between inflation expectations and spending has been examined empirically by Coibion et al. (2022), who find that households act on their inflation expectations when making spending and saving decisions.

This empirical regularity may also be examined annually in the BoE IAS. Question 17 asks which actions households are planning to take in response to their twelve-month-ahead inflation expectations. Among the potential responses, I focus on the two options related to consumption behavior: bringing forward major purchases (the intertemporal-substitution response) and cutting back spending to save more. Figure A1 depicts the weighted share of each cohort selecting each intended action. Importantly, both margins move with the inflation cycle: the shares fall through the low-inflation 2010s and rise sharply during the 2021–2023 surge. The responses are also heterogeneous across cohorts: during episodes of higher inflation, younger and middle-aged cohorts tend to report higher intentions to adjust consumption on both margins than older cohorts do.

FIGURE A1. **Consumption Responses to Inflation Expectations**



A. Bring forward major purchases.

B. Cut back spending and save more.

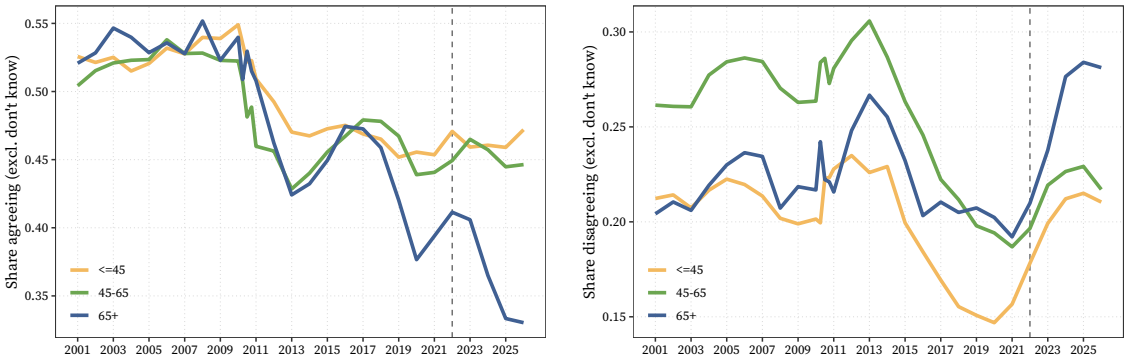
Note: Weighted share of respondents in each cohort selecting the action in BoE IAS question Q17.

Fact A2. The Perceived Channel.

"Older cohorts that have lived through higher inflationary episodes perceive higher inflation. Besides, since the 2021–2022 inflationary shock, they have become the less convinced that higher interest rates restrain price increases."

The final fact provides evidence on the influence of households perception about the transmission of monetary policy and their perception of the inflation rate. To document the change in perceived transmission, the survey asks whether households agree with the following statement: "A rise in interest rates would make prices in the high street rise more slowly in the medium term (say a year or two)." Panel A2A plots the weighted share who answered "agree" or "strongly agree," while Panel A2B plots the share who answered "disagree" or "strongly disagree." Both panels mirror each other. Before 2022, the proportion of households reporting agreement or disagreement moves in parallel across cohorts. However, after the 2021–2022 inflationary shock, the oldest cohort reports more scepticism about whether higher rates would reduce inflation.

FIGURE A2. **Perceived Monetary Transmission by Cohort**

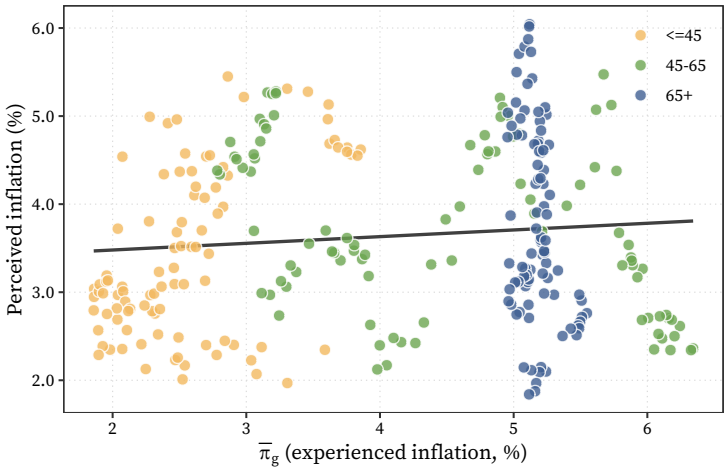


A. Share agreeing. B. Share disagreeing.

Note: Asked in question Q9b. Both shares exclude "don't know" from the denominator. Included waves 2001–2026.

Next, turning to perceived inflation, Question 1 of the survey asks about the households perception of how prices have changed over the last twelve months. Figure A3 plots the inflation experiences against the perceived inflation rate for each cohort and reveals that older cohorts with a higher experienced-inflation report higher perceived inflation. This result aligns with Malmendier and Nagel (2016), in the sense that the realized inflation history a cohort has lived through shapes how it perceives current prices. It differs, however, from attention-based models of perceived inflation such as Francis (2026), in which perceptions diverge from official inflation because households overweight salient or volatile goods.

FIGURE A3. **Perceived Inflation and Inflation Memories**



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